

SS1 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

FIRST TERM

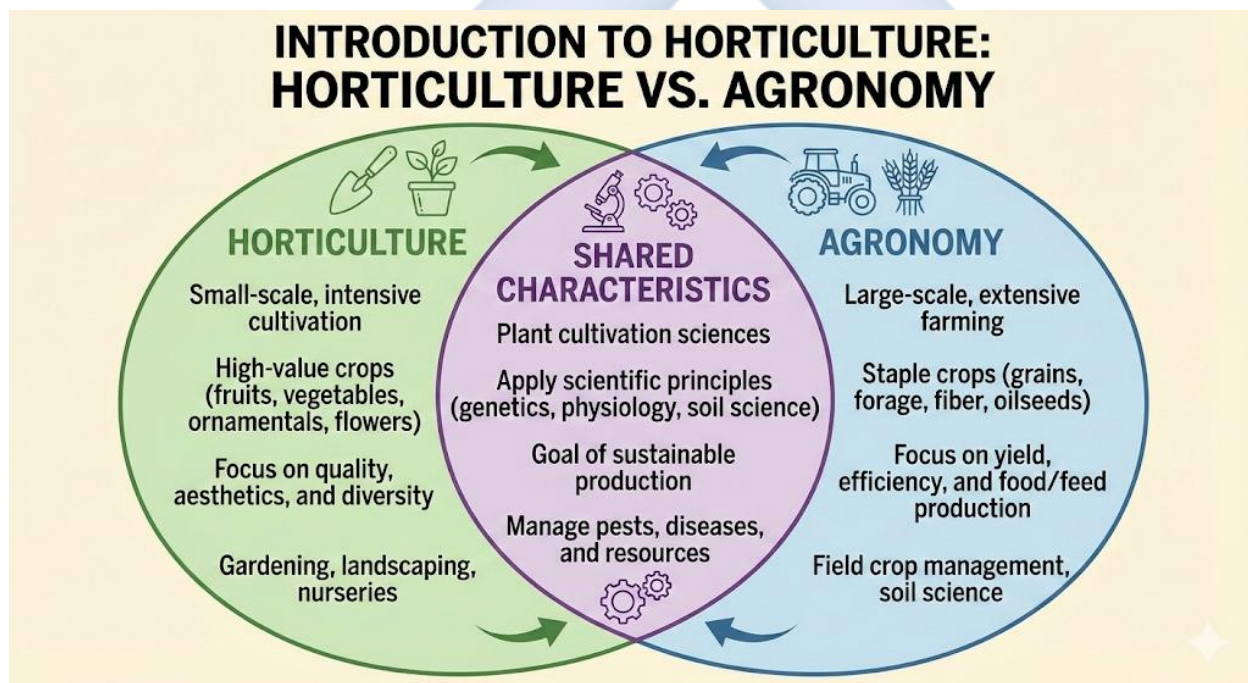
WEEK 1: INTRODUCTION TO HORTICULTURE

✿ MEANING OF HORTICULTURE

The word "**horticulture**" is derived from two Latin words:

- *hortus*, which means "garden"
- *cultura*, which means "cultivation"

Therefore, horticulture literally means "**garden cultivation.**"



Horticulture is the specialized branch of agricultural science that deals with the **intensive** production, processing, and marketing of high-value plants, primarily fruits, vegetables, flowers, and ornamental plants.

It is crucial to distinguish horticulture from its related field, **agronomy**:

Feature	HORTICULTURE	AGRONOMY
Definition	"Garden cultivation"	"Field cultivation"
Scale	Intensive (high input, high value per plant/area)	Extensive (large scale, e.g., hectares)
Crops	Fruits, vegetables, flowers, herbs, ornamental plants	Grains, cereals, tubers, fibers (staple/field crops)
Examples	Tomatoes, Mangoes, Okra, Roses, Carpet Grass	Maize (Corn), Rice, Cassava, Soybeans, Cotton
Goal	High quality, aesthetics, and high value	High yield (bulk) and long-term storage

SCOPE OF HORTICULTURE

The "scope" of horticulture refers to the wide range of activities and industries it covers. It is not just about farming; it is a complete "seed-to-consumer" chain.

1. **Production:** The primary cultivation of horticultural crops (fruits, vegetables, etc.).
2. **Nursery Management:** The specialized production of young plants, seedlings, and "saplings" (young trees) for sale to other farmers or landscapers.
3. **Processing (Value Addition):** Transforming raw horticultural products into more stable and valuable goods.
 - *Examples:* Juicing oranges, canning tomatoes into paste, drying herbs, extracting essential oils from flowers, freezing vegetables.
4. **Marketing and Export:** The business of selling fresh and processed products. This includes local markets, supermarkets, and the international export of high-value crops (like ginger, hibiscus, or cashew).
5. **Landscaping and Garden Design (Ornamental Horticulture):** The art and science of designing, installing, and maintaining beautiful, functional outdoor spaces (e.g., parks, home gardens, office complexes, roadside beautification).

6. **Research and Development:** Breeding new varieties of plants that are higher-yielding, more colorful, or more resistant to diseases.

IMPORTANCE OF HORTICULTURE IN NIGERIA

Horticulture is a cornerstone of Nigeria's food security, economy, and environmental health.

1. NUTRITIONAL IMPORTANCE (FOOD SECURITY):

- **Main Source of Vitamins and Minerals:** Horticultural crops are the primary source of essential vitamins (like Vitamin C from citrus, Vitamin A from carrots and peppers) and minerals in the Nigerian diet.
- **Dietary Variety:** They provide the essential "soup" and "stew" ingredients (tomatoes, peppers, onions, okra, *ugu*, *ewedu*) that accompany staple foods (like rice, eba, fufu), making them palatable and nutritious.
- **Prevents Malnutrition:** A diet rich in fruits and vegetables prevents "hidden hunger" and deficiency diseases like scurvy (lack of Vit. C) and night blindness (lack of Vit. A).

2. ECONOMIC IMPORTANCE:

- **Massive Job Creation:** It is highly **labor-intensive**, creating millions of jobs for farmers, nursery workers, landscapers, fruit pickers, marketers (e.g., "Mama Put," tomato sellers), and transport workers.
- **High Income for Farmers:** Horticultural crops (like tomatoes, peppers) often have a *much higher market value* per hectare than agronomic crops (like maize).
- **Raw Materials for Industry:** It supplies the raw materials for Nigeria's processing industries (e.g., fruit juice companies, tomato paste factories, canning companies).
- **Foreign Exchange (Export):** Nigeria is a major exporter of horticultural products like hibiscus (*zobo* flower), ginger, garlic, cashew, and mangoes, which brings in valuable foreign currency.

3. AESTHETIC AND ENVIRONMENTAL IMPORTANCE:

- **Beautification (Landscaping):** Ornamental horticulture (planting flowers, trees, and lawns) beautifies our homes, cities, roads, and parks.
- **Environmental Health:**
 - **Shade and Cooling:** Trees in urban areas reduce the "heat island" effect, providing shade and cooling the air.
 - **Erosion Control:** Planting groundcovers and trees helps control soil erosion, especially in erosion-prone areas of the Southeast.
 - **Air Quality:** Plants absorb carbon dioxide (CO_2) and release oxygen (O_2).

4. MEDICINAL IMPORTANCE:

- Many horticultural plants are the foundation of traditional medicine and modern pharmaceuticals (e.g., neem (*dogonyaro*), garlic, ginger, lemongrass, bitter leaf).

EVALUATION

1. What is the literal meaning of the word "horticulture," based on its Latin roots?
2. In a table, state three key differences between **horticulture** and **agronomy**.
3. What is "value addition" in horticulture? Give two examples.
4. Explain two economic and two nutritional importances of horticulture in Nigeria.
5. What is "hidden hunger," and how does horticulture help prevent it?

ASSIGNMENT

1. Write a 250-word essay on the topic: "Why Horticulture is More Than Just Farming."
2. Survey your local market. List five (5) horticultural crops you see for sale and five (5) agronomic crops.
3. Research one Nigerian horticultural export crop (e.g., Hibiscus/Zobo, Ginger, or Cashew). Write a one-page report on its production and economic value to Nigeria.
4. Define the following terms: Intensive cultivation, Aesthetics, GDP, Erosion, Deficiency Disease.
5. Interview a local garden owner, landscaper, or "Mama Put" vendor. Ask them why the crops they use are important. Write a short summary of their response.

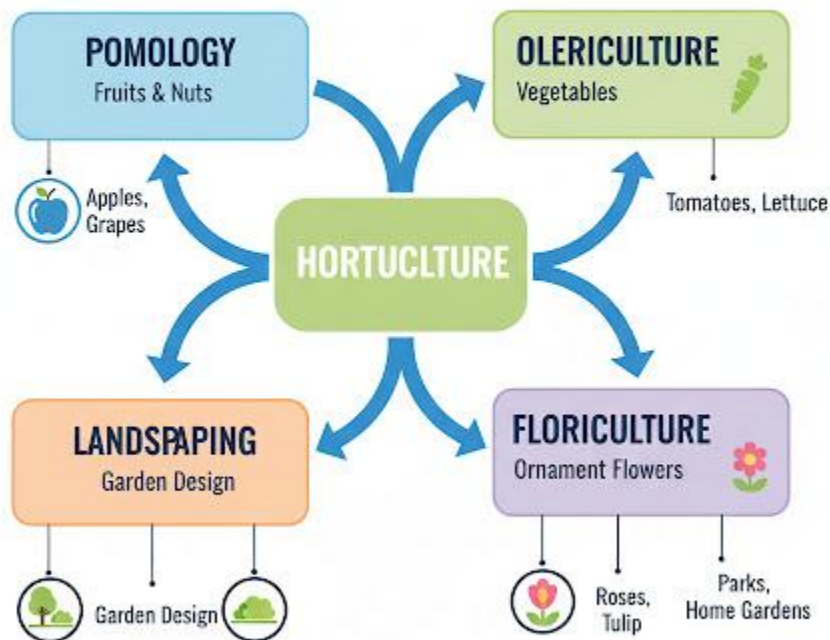


WEEK 2: BRANCHES OF HORTICULTURE

🌱 INTRODUCTION

Horticulture is a massive field. To study it properly, it is broken down into four major specialized branches, each with its own set of crops and practices.

BRANCHES OF HORTICULTURE



1. POMOLOGY (FRUIT PRODUCTION)

- **Definition:** **Pomology** is the branch of horticulture that deals with the science, cultivation, processing, and marketing of **fruit** crops. The name comes from the Latin word ***pomum***, meaning "fruit."
- **Scope:** This branch focuses on **perennial** plants (plants that live for many years). It is a long-term investment.

- **Sub-categories and Examples:**
 - **Tree Fruits:**
 - *Tropical:* **Mango, Cashew, Avocado, Guava, Oil Palm, Cocoa, Kola nut, Coconut.**
 - *Citrus:* **Orange, Lemon, Lime, Grapefruit.**
 - *Temperate:* (Grown in high-altitude areas like Jos Plateau) **Apple, Pear, Plum.**
 - **Small Fruits & Vines:**
 - **Pineapple, Banana, Plantain, Pawpaw (Papaya), Grapes.**
- **Note:** Pomology also includes the cultivation of **nuts** (like Cashew) and "industrial" fruits (like **Oil Palm** and **Cocoa**).

2. OLERICULTURE (VEGETABLE PRODUCTION)

- **Definition:** Olericulture is the branch of horticulture that deals with the science, cultivation, processing, and marketing of **vegetables**. The name comes from the Latin word *oleris*, meaning "pot-herb."
- **Scope:** This branch focuses primarily on **annual** plants (plants that complete their life cycle in one season). It provides the quickest return on investment.
- **Sub-categories (Classification by Use):**
 - **Leafy Greens:** These are grown for their leaves.
 - *Examples:* **Ugu (Fluted Pumpkin), Spinach (Efo, Amunututu), Ewedu, Lettuce, Cabbage, Bitter Leaf.**
 - **Fruit Vegetables:** These are botanically "fruits" but are eaten as vegetables.
 - *Examples:* **Tomato, Pepper (all kinds), Okra, Eggplant, Cucumber, Watermelon, Melon (Egusi).**
 - **Root, Tuber, & Bulb:** These are grown for their underground storage organs.
 - *Examples:* **Carrot, Onion, Garlic, Radish, Beetroot.** (Note: Yams and Cassava are usually considered agronomy, but small-scale vegetable yams can be olericulture).

- **Legumes (Pulses):** Grown for their edible pods or seeds.
 - *Examples:* **Green Beans, Peas.**

3. FLORICULTURE (FLOWER & ORNAMENTAL PRODUCTION)

- **Definition: Floriculture** is the branch of horticulture that deals with the cultivation, processing, and marketing of **flowers** and **ornamental plants**. The name comes from the Latin word *flos*, meaning "flower."
- **Scope:** This is a high-value industry focused purely on **aesthetics (beauty)**.
- **Sub-categories:**
 - **Cut Flowers:** This is the production of flowers (stems, heads) that are cut from the plant for use in bouquets, wreaths, and arrangements.
 - *Examples:* **Rose, Marigold, Lily, Sunflower, Heliconia.**
 - *Note:* This is a major industry in Jos, Plateau State, due to the cool climate.
 - **Potted Plants (Pot Plants):** This is the production of ornamental plants (flowering or foliage) grown in containers for indoor or outdoor decoration.
 - *Examples:* **Hibiscus, Croton, Ferns, Orchids.**
 - **Bedding Plants:** These are small, flowering plants (like Marigolds) grown in bulk for mass planting in garden beds and parks.

4. LANDSCAPING (ORNAMENTAL HORTICULTURE)

- **Definition: Landscaping** (or Landscape Horticulture) is the broadest branch. It is the **art and science** of **designing, installing, and maintaining** attractive and functional outdoor spaces.
- **Scope:** It is the "application" branch that *uses* the products from Floriculture and Pomology (ornamental trees) and combines them with non-living elements (**hardscapes**).
- **Key Components:**
 - **Design:** The "blueprint" or "plan" for the garden.
 - **Installation:** The physical work of planting and building.

- **Maintenance:** The ongoing work of weeding, pruning, and mowing.
- **Turf Management:** The specialized cultivation of lawn grasses (e.g., **Carpet Grass, Augustine Grass**).
- **Nursery Management:** Growing the trees, shrubs, and groundcovers used in the designs.
- **Arboriculture:** The specialized study and care (pruning, felling, treatment) of individual, large trees (*arbor* = tree).
- **Application:** Public parks, private gardens, office complexes, university campuses, roadside beautification, golf courses.

EVALUATION

1. What is **Pomology**? Give one example of a Tree Fruit and one example of a Small Fruit.
2. What is **Olericulture**? List the 4 main sub-categories of vegetables.
3. A farmer grows *Ugu* (Fluted Pumpkin). Which branch of horticulture are they practicing?
4. What is the main difference between **Floriculture** and **Landscaping**?
5. What is "Turf Management," and which branch does it fall under?

ASSIGNMENT

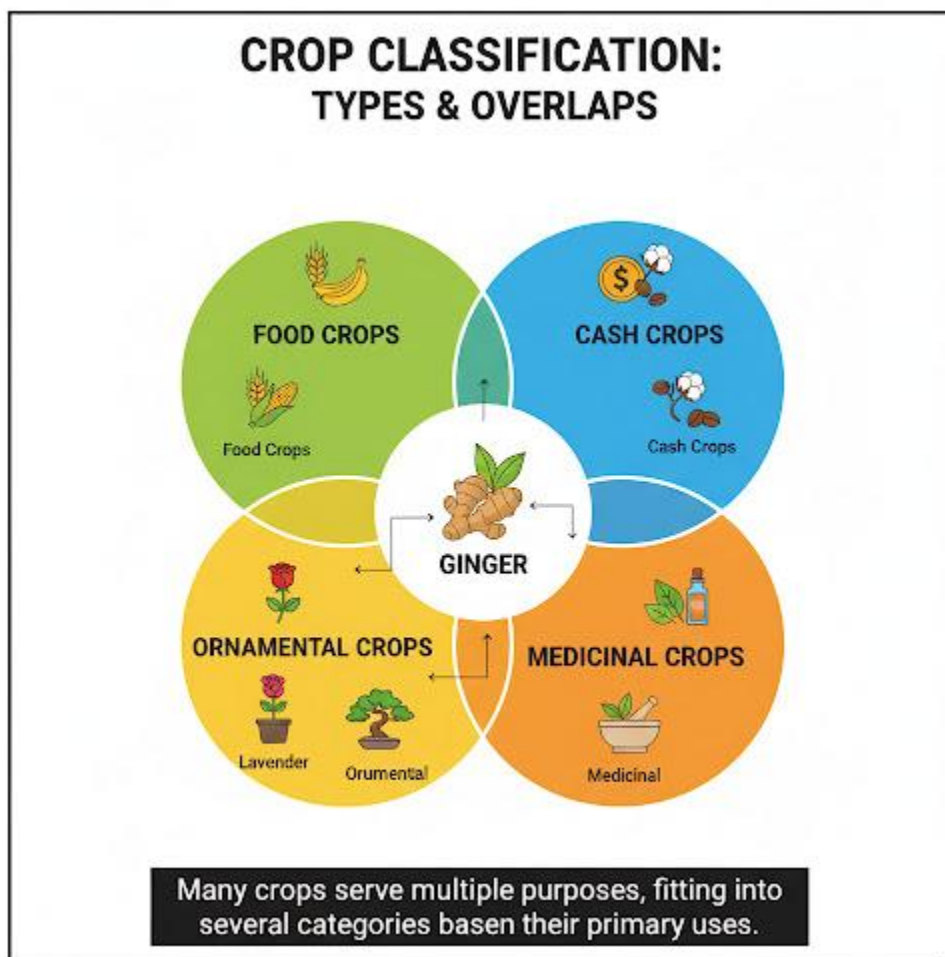
1. Write a 250-word essay on the branch of horticulture that interests you the most, explaining your choice.
2. Draw a simple design for a small "home garden," showing where you would place **Pomology** (e.g., one mango tree), **Olericulture** (a small vegetable bed), and **Floriculture** (a flower border).
3. What is a "Cut Flower"? Why is this industry well-suited to the Jos Plateau?
4. Define the following terms: Perennial, Annual, Aesthetics, Hardscape, Arboriculture.
5. Create a table that classifies the following 8 crops into their correct branch (Pomology, Olericulture, or Floriculture): **Tomato, Mango, Rose, Cabbage, Pineapple, Hibiscus, Onion, Orange**.

WEEK 3: CROP CLASSIFICATION

INTRODUCTION

In Week 2, we classified the *branches of horticulture*. This week, we classify the crops themselves based on their **primary use** in human society and the economy.

It is important to note that these categories are not always exclusive. A single crop can easily fall into **multiple categories** at the same time.



1. FOOD CROPS

- **Definition:** These are crops that are grown for the primary purpose of being **eaten by humans** (or fed to livestock). They form the foundation of our food security.
- **Sub-categories:**

- **Staple Foods (Agronomic):** These are the high-energy, bulk crops that form the *base* of our diet (e.g., Rice, Maize, Cassava, Yam, Millet).
- **Horticultural Food Crops:** These are the crops that *supplement* the staples, providing essential vitamins, minerals, and flavors.
- **Horticultural Examples:**
 - **Vegetables:** Tomatoes, Onions, Peppers, Okra, Cabbage, *Ugu*, *Amunututu* (Spinach).
 - **Fruits:** Oranges, Mangoes, Pawpaw, Bananas, Pineapples, Plantains.

2. CASH CROPS

- **Definition:** A **cash crop** is any crop that is grown primarily for **sale (commerce)** to make a **profit**.
- **The Concept:** They are not grown for the farmer's own family to eat (subsistence). They are grown to be *sold*, either to a **local market**, an **industry (processor)**, or for **export**.
- **The Overlap:** *Any* crop can be a cash crop. A tomato farmer in Kano growing 20 hectares of tomatoes is growing a **food crop** that is *also* his **cash crop**. "Cash crop" simply implies the primary motive is profit.
- **Traditional Cash Crops (Mainly for Export):**
 - **Cocoa** (A fruit crop, pomology): The beans are exported to make chocolate. This is Nigeria's *largest* agricultural export.
 - **Oil Palm** (A fruit crop, pomology): The fruit is processed into palm oil (for food) and palm kernel oil (for cosmetics and industry).
 - **Cashew** (A fruit crop, pomology): The nuts are exported.
 - **Rubber** (A tree crop, non-food): The latex is "tapped" for the tire and manufacturing industry.
- **Horticultural Cash Crops:**
 - **Hibiscus (Zobo)** (A flower crop, floriculture): The dried calyx is exported in massive quantities (especially to Mexico).
 - **Ginger** (A vegetable, olericulture): Exported as a spice.

- **Sesame Seeds**
- **Garlic**

3. ORNAMENTAL CROPS

- **Definition:** Crops that are grown *not* for food or medicine, but purely for their **beauty (aesthetic value)**. They are the primary crops of the **Floriculture** and **Landscaping** branches.
- **Purpose:** Used for interior and exterior decoration, beautification, shade, and landscaping.
- **Horticultural Examples:**
 - **Turf (Grass):** Carpet Grass, Augustine Grass (used for lawns).
 - **Flowering Shrubs:** **Hibiscus** (ornamental varieties), *Ixora* ("Pride of Barbados"), Allamanda, Bougainvillea, Roses.
 - **Ornamental Trees:** **Queen's Palm, Royal Palm**, Masquerade Tree (Vertical Tree), *Delonix regia* (Flame Tree).
 - **Groundcovers:** Plants used to cover the soil (e.g., "Step-light").

4. MEDICINAL CROPS

- **Definition:** Crops that are cultivated for their **therapeutic properties** or as the source of ingredients for **traditional medicine** ("herbs") or modern **pharmaceuticals**.
- **The Overlap:** Many of these are also food crops, but they are consumed for their health benefits.
- **Horticultural Examples:**
 - **Ginger** (*Zingiber officinale*): Used as an anti-inflammatory and for treating nausea, colds, and coughs.
 - **Garlic** (*Allium sativum*): Used for its antimicrobial (antibacterial) properties and to manage high blood pressure.

- **Neem Tree** (*Azadirachta indica* / *Dogonyaro*): This is a "miracle" tree. The leaves, bark, and seeds are used to make antiseptic soaps and teas (traditionally for malaria).
- **Aloe Vera**: The gel from the leaves is a common horticultural plant, used to treat burns, wounds, and skin irritations.
- **Lemongrass** (*Cymbopogon*): The leaves are boiled for a fever-reducing tea.
- **Bitter Leaf** (*Vernonia amygdalina* / *Ewuro*): The "washed" juice is used as a tonic to cleanse the blood and for its anti-diabetic and antimicrobial properties.
- **Pawpaw (Papaya)**: The leaves and unripe fruit contain *Papain*, an enzyme that aids digestion.

EVALUATION

1. What is the primary basis for the crop classification learned this week (e.g., is it botany, life cycle, or use)?
2. Define a "Cash Crop." Give two examples of traditional cash crops.
3. Explain the concept of "overlap." Give an example of one crop that is a **Food Crop**, a **Cash Crop**, AND a **Medicinal Crop**.
4. What is the primary purpose of growing **Ornamental Crops**?
5. List three medicinal crops commonly used in Nigeria and their traditional use.

ASSIGNMENT

1. Write a 250-word essay on the topic: "The Economic Importance of Cash Crops to Nigeria." You must mention Cocoa and Oil Palm.
2. Create a "local pharmacy" poster. Draw or find pictures of 5 medicinal plants, label them (e.g., "Bitter Leaf," "Neem"), and write their traditional use.
3. Survey your school compound or home. List 5 different *ornamental* plants that you can see.
4. What is the difference between a staple food crop and a horticultural food crop?
5. Define: Subsistence, Export, Aesthetics, Therapeutic, Pharmaceutical.



WEEK 4: TOOLS AND EQUIPMENT

✂ INTRODUCTION

Tools and equipment are the physical implements used to carry out all horticultural operations. They are the "hands" of the farmer. Proper **identification**, **use**, and **maintenance** are essential for **efficiency** (doing the job well), **safety** (preventing injury), and the **longevity** of the tools (saving money).

They can be classified by their primary function:



1. CUTTING TOOLS

- **CUTLASS (MACHETE):**
 - **Identification:** A long, heavy, single-edged blade with a wooden or plastic handle. It is the most common tool in West Africa.
 - **Uses:** A multi-purpose, heavy-duty tool. Used for:
 - **Clearing:** Cutting thick bushes, grasses, and small trees.
 - **Weeding:** General weeding in a large area.
 - **Harvesting:** Harvesting large crops like Cassava, Sugarcane, or Plantain.

- **Care:** Sharpen the blade regularly with a **flat file**. Keep the blade clean and dry (or lightly oiled) to prevent rust. Store in a sheath.
- **SECATEURS (HAND PRUNERS / PRUNING SHEARS):**
 - **Identification:** A small, one-handed, scissor-like tool with two sharp, curved blades.
 - **Uses:** For *precise* and *clean* cutting, which is vital in horticulture.
 - **Pruning:** Removing small, diseased, or unwanted branches from fruit trees and shrubs.
 - **Harvesting:** Harvesting flowers or fruits (like tomatoes, peppers) with a clean cut.
 - **Care:** Clean sap and dirt from the blades after *every* use. Oil the hinge/pivot point. Keep sharp.
- **SHEARS (HEDGE SHEARS):**
 - **Identification:** A large, two-handed tool with long, flat blades (like giant scissors).
 - **Uses:** For **trimming and shaping hedges**, lawns, and ornamental shrubs. It is used for *volume* trimming, not precise cuts.
 - **Care:** Keep blades clean and sharp. Oil the hinge.
- **BUDDING / GRAFTING KNIFE:**
 - **Identification:** A small, *extremely sharp* pocket knife, often with a special "bark-lifter" on the back of the blade.
 - **Uses:** Exclusively for **crop propagation** (Week 9). Used for making the clean, surgical cuts needed for grafting and budding.
 - **Care:** Must be kept **sterile** (e.g., cleaned with alcohol) when in use to prevent transmitting diseases between plants.

2. DIGGING AND SOIL CULTIVATION TOOLS

- **SPADE:**
 - **Identification:** Has a **flat, rectangular blade** with a **straight** bottom edge.

- **Uses:** For *cutting* clean, straight-sided holes. Ideal for digging planting holes, "edging" lawns, and cutting through tough soil and roots.
- **SHOVEL:**
 - **Identification:** Has a **curved blade** with a **rounded or pointed tip**.
 - **Uses:** For *moving* loose materials. It is a "scooping" tool. Used for moving soil, sand, compost, manure, or sand.
 - (*Spade = digging, Shovel = moving*)
- **GARDEN FORK (DIGGING FORK):**
 - **Identification:** A fork with 4-6 strong, thick tines (prongs).
 - **Uses:**
 - For **aerating** (loosening) heavy, **compacted, or clayey soil** (where a spade is hard to use).
 - For turning compost heaps.
 - For harvesting root crops (like Irish Potatoes or Carrots) without damaging them.
- **HOE (WEST AFRICAN HOE):**
 - **Identification:** A heavy-duty blade set at an angle to a (usually wooden) handle.
 - **Uses:** The primary tool for:
 - **Weeding** (cutting weeds at the soil line).
 - Making **ridges, heaps (mounds)**, or seedbeds.
 - Breaking up large clods of soil.
- **HAND TROWEL:**
 - **Identification:** A small, handheld, scoop-shaped tool.
 - **Uses:**
 - **Transplanting** seedlings from the nursery (Week 8) to the field.
 - Planting in pots and containers.
 - Small-scale weeding in flower beds.
- **RAKE:**
 - **Identification:** A long handle with a "comb" of tines (metal or plastic) at the end.

- **Uses:**
 - Creating a **fine, level seedbed** after digging.
 - Removing stones, debris, and pulling up shallow-rooted weeds.
 - Gathering dry leaves or cut grass.

3. WATERING AND APPLICATION EQUIPMENT

- **WATERING CAN:**

- **Identification:** A container with a handle and a spout.
- **Key Feature:** Must have a detachable "**rose**" (the sprinkler head).
- **Uses:** For watering seedlings in the nursery and potted plants. The "rose" breaks the heavy stream of water into gentle droplets that *will not* dislodge the small seedlings.

- **KNAPSACK SPRAYER:**

- **Identification:** A tank (15-20L) worn on the back, a hand-pump lever, a hose, and a long "wand" with a nozzle.
- **Uses:** For spraying liquid chemicals (**pesticides, herbicides, fungicides**) or liquid fertilizer (**foliar application**).
- **Care (CRITICAL):**
 - i. Wash *thoroughly* after every use with soap and water.
 - ii. **NEVER** use the same sprayer for **herbicides** (weed killers) and **pesticides/fertilizers**. The residue from the herbicide *will* kill your crops. You must have two separate, labeled sprayers.

- **WHEELBARROW:**

- **Identification:** A single-wheeled cart with two handles.
- **Uses:** For transporting heavy or bulky materials (compost, manure, soil, harvested crops, tools).

EVALUATION

1. What is the main difference in *use* between a **Spade** and a **Shovel**?
2. What is a **Secateur** used for, and why is it better than a cutlass for this job?
3. What is the "rose" on a watering can, and why is it essential for nursery work?
4. What is the *most important* maintenance rule for a **Knapsack Sprayer**?
5. What tool is best for digging in heavy, compacted clay soil?

ASSIGNMENT

1. Write a one-page "Tool Maintenance Guide." Describe the 4 key maintenance steps that apply to *all* metal tools (e.g., cleaning, drying, sharpening, storing).
2. Draw and label the following 5 tools: **Cutlass, Secateurs, Spade, Hand Trowel, Watering Can (with rose).**
3. Visit a gardener or farmer (or your school's store). Identify 5 tools they use. Write down the *local name* and the *English name*.
4. What is the difference in use between a "Secateur" and a "Hedge Shear"?
5. Define: Tines, Sheath, Aeration, Hinge, Transplanting, Herbicide.

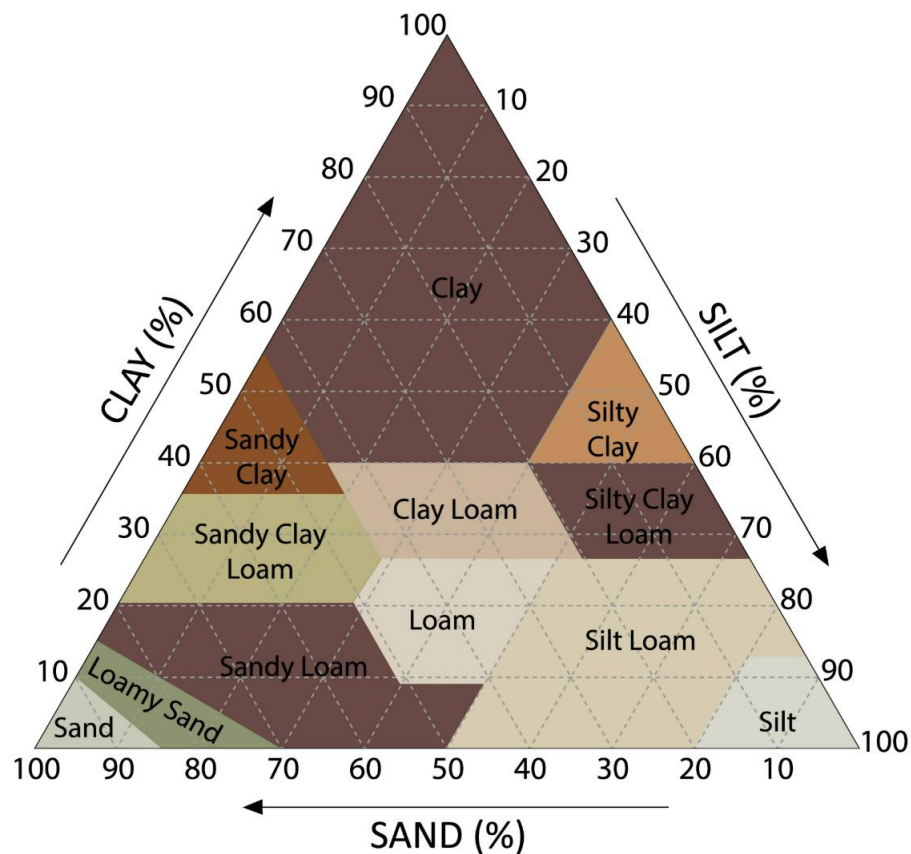
WEEK 6: SOIL IN HORTICULTURE

🌱 INTRODUCTION TO HORTICULTURAL SOIL

Soil is the foundation of all terrestrial plant life. It is *not* just "dirt." It is a complex, living ecosystem made of four main components:

1. **Mineral Particles** (sand, silt, clay) - $\approx 45\%$
2. **Organic Matter (Humus)** (dead/decaying plants and animals) - $\approx 5\%$
3. **Water** (held in pore spaces) - $\approx 25\%$
4. **Air** (held in pore spaces) - $\approx 25\%$

For horticulture, the soil must be of a *higher quality* (a "loam") than for agronomy because horticultural crops are generally more "needy" and have shallower root systems.



SOIL TYPES (CLASSIFIED BY TEXTURE)

Soil texture refers to the *percentage* (the relative proportion) of **sand**, **silt**, and **clay** particles in the soil. This is the most important physical property, as it determines water retention and drainage.

1. SANDY SOIL:

- **Identification:** Large particles, feels **gritty** and coarse. It does not stick together.
- **Properties:**
 - **Pros:** Drains *very fast* (good aeration), heats up quickly in the spring.
 - **Cons:** **Poor water and nutrient retention.** Water and fertilizer "leach" (are washed) straight through. It is prone to drought. It is a "light soil."

2. CLAYEY SOIL:

- **Identification:** Very small, fine particles. Feels **sticky** when wet and forms **hard, brick-like clods** when dry.
- **Properties:**
 - **Pros:** **Excellent water and nutrient retention.** It holds onto water and fertilizer for a long time.
 - **Cons:** **Poor drainage** (leads to **waterlogging**), poor aeration (plant roots can suffocate), and it is very *hard to dig* (a "heavy soil").

3. LOAMY SOIL (THE IDEAL SOIL):

- **Definition:** This is the **perfect** soil for horticulture. It is *not* a "pure" soil, but a *balanced mixture* of sand, silt, and clay (e.g., 40% Sand, 40% Silt, 20% Clay).
- **Identification:** Feels soft, crumbly, and slightly moist.
- **Properties:** It has the "**best of all worlds**" — it **holds water and nutrients** (from the clay/silt) but also **drains well and has good aeration** (from the sand). It is easy to work with.

PROPERTIES OF SOIL

- **SOIL STRUCTURE:** This is how the sand, silt, and clay particles *clump together* (aggregate) to form "peds." A good, "crumbly" or "granular" structure is essential for water and air to penetrate.
- **SOIL pH:**
 - **Definition:** A measure of the soil's **acidity** or **alkalinity**, on a scale of 0-14.
 - **pH 7 = Neutral**
 - **pH < 7 = Acidic** (or "sour")
 - **pH > 7 = Alkaline** (or "basic" / "sweet")
 - **Importance:** Soil pH is the "master switch" that controls **nutrient availability**. Most horticultural crops (vegetables, fruits) grow best in **slightly acidic to neutral soil (pH 6.0 - 7.0)**.
 - **The Problem:** If the soil is *too* acidic (e.g., pH 4.5, common in the high-rainfall South), essential nutrients like Phosphorus and Calcium become "locked" and unavailable to the plant, even if they are in the soil.
- **SOIL ORGANIC MATTER (HUMUS):** This is the "secret ingredient" to great soil. It is the dark, crumbly, nutrient-rich material that results from the decomposition of plants and manure.

SOIL MANAGEMENT TECHNIQUES

These are the practices used to *improve* the soil to make it more "loamy" and fertile.

1. MANURING / COMPOSTING (ADDING ORGANIC MATTER):

- **Definition:** The addition of **organic matter** (like cow dung, poultry droppings, or compost) to the soil.
- **This is the single most important management practice.**
- **Benefits:**
 - It feeds the soil microbes (the "life" in the soil).
 - It provides essential nutrients.

- It **improves clay soil** by "fluffing it up" (creating aggregates) and improving drainage.
- It **improves sandy soil** by acting like a "sponge" and helping it *hold* water and nutrients.

2. LIMING:

- **Definition:** The application of **lime** (which is calcium carbonate, e.g., from ground limestone or chalk) to the soil.
- **Purpose:** To **raise the pH** of soil that is **too acidic**. This "sweetens" the soil and "unlocks" the nutrients.

3. MULCHING:

- **Definition:** Covering the surface of the soil with a layer of material.
- **Types of Mulch:**
 - *Organic:* Dry leaves, straw, wood chips, grass clippings.
 - *Synthetic:* Black plastic polythene sheets.
- **Benefits:**
 - **Conserves moisture** by reducing evaporation from the sun.
 - **Suppresses weeds** by blocking sunlight.
 - **Regulates soil temperature** (keeps it cooler in the day and warmer at night).
 - Organic mulch adds nutrients as it decomposes.

4. CROP ROTATION:

- **Definition:** The practice of *not* planting the same type of crop in the same spot, year after year. (e.g., Year 1: Tomato; Year 2: Beans; Year 3: Cabbage).
- **Benefits:**
 - **Pest/Disease Control:** Pests that attack tomatoes will die off if you plant beans there next.
 - **Nutrient Management:** Different crops take different nutrients. (Legumes like beans actually *add* nitrogen to the soil).

EVALUATION

1. What are the four main components of soil?
2. Describe the three main soil types (Sand, Silt, Clay) based on their *feel* and *water-holding capacity*.
3. What is **Loamy soil**, and why is it considered ideal?
4. What is **Soil pH**, and why is it the "master switch" for nutrient availability?
5. What is **Liming**, and what specific problem does it solve?

ASSIGNMENT

1. Write a 250-word essay on **Manuring/Composting**. Your essay must explain how this one practice can fix the problems of *both* sandy soil *and* clay soil.
2. What is **Mulching**? List two *organic* mulching materials and one *synthetic* one.
3. Draw a simple "Crop Rotation" plan for a 3-year cycle on one plot of land. (e.g., Year 1: Maize, Year 2: Beans, Year 3: Okra).
4. Define the following terms: Humus, Leaching, Waterlogging, pH, Nutrient Availability.
5. **Practical:** Collect a soil sample from your home or school. (a) Rub it between your wet fingers. Is it gritty (Sandy) or sticky (Clayey)? (b) Place it in a clear bottle, fill with water, shake, and let it settle for 30 minutes. Draw the layers you see (sand should be at the bottom).

WEEK 8: NURSERY PRACTICES

🌱 INTRODUCTION TO THE NURSERY



A **nursery** is a special, dedicated plot of land (or facility) where young, tender plants (seedlings) are raised with **intensive care** before being moved to the permanent field.

Why use a nursery?

1. **Protection:** Seedlings are extremely vulnerable. A nursery protects them from harsh sun, heavy rain, strong winds, pests (birds, insects), and diseases.
2. **Efficiency:** It is far easier to water, weed, and manage 1,000 seedlings in one small, 1m x 5m bed than it is to manage them scattered across a giant 1-hectare field.
3. **Selection:** It allows the farmer to select only the *strongest* and *healthiest* seedlings for planting in the field. This ensures a uniform, high-quality crop.
4. **Saves Time:** The farmer can be raising seedlings in the nursery *at the same time* as they are preparing the main field (e.g., ploughing, ridging).

Crops that require a nursery: Almost all small-seeded vegetables (e.g., **Tomato, Pepper, Onion, Cabbage, Lettuce, Eggplant**) and many tree crops (e.g., **Mango, Cocoa, Oil Palm**).

SEED SELECTION

Good crops start with good seeds. A good seed must have:

- **Viability:** The seed must be **viable** (alive and capable of germinating). You can test this with a **germination test** (e.g., plant 10 seeds in a wet cloth; if 9 sprout, you have 90% viability).
- **Purity:** The seed must be "true to type," meaning it is the correct variety (e.g., "UC-82b Tomato") and is free from weed seeds or other crop seeds.
- **Health:** The seed must be mature, fully-formed, and free from disease, insect damage, or cracks.

GERMINATION AND SEED DORMANCY

Germination is the process where the seed embryo "wakes up," breaks its coat, and sprouts (develops a root and a shoot). **Seed Dormancy** is a "sleeping" state where a viable (living) seed *will not* germinate, even if conditions (water, oxygen, warmth) are perfect. The seed is "waiting" for a special signal. We must "break" this dormancy.

- **Methods to Break Dormancy (Pre-treatment):**
 - a. **Scarification (Physical):** Breaking, scratching, or softening the *hard, waterproof seed coat*. This is for seeds with very hard shells (e.g., *Agbalumo*, Mango, Locust bean, some nuts).
 - **Methods:** Filing the seed with a metal file, rubbing with sandpaper, or **soaking in hot water** (pour boiling water over them and let them soak as it cools).
 - b. **Soaking:** Soaking seeds in regular water for 12-24 hours can soften the coat and speed up germination for many vegetables.

NURSERY PRACTICES (THE "HOW-TO")

1. **Site Selection:** The nursery must be:

- On **level ground** (to prevent water from washing seeds away).
- Near a reliable **water source** (critical).
- In a **protected** area (e.g., away from strong winds, secure from animals).

2. Nursery Beds:

- **Raised Beds:** Soil is heaped up (15-20cm high). This is the *best* method for the rainy season as it provides **excellent drainage** and prevents seedlings from drowning.
- **Sunken Beds:** Soil is dug *down* 15-20cm. This is used in very hot, dry (Sahel) regions to *conserve* water.
- **Containers:** Sowing seeds in **nursery trays** (plug trays), **polythene bags** (polybags), or **pots**. This is the *best* method for delicate or high-value crops (like tree crops) as it *does not disturb the roots* during transplanting.

3. Sowing Seeds:

- **Broadcasting:** Scattering the seeds randomly over the bed. This is wasteful and leads to overcrowding and difficult weeding.
- **Drilling:** Sowing the seeds in small, shallow lines (drills). This is the *preferred* method. It uses less seed, and makes weeding, thinning, and chemical application much easier.

4. Nursery Management (After Sowing):

- **Watering:** Must be done *gently* (with a **watering can + rose**) 1-2 times per day. The soil must be kept moist, not waterlogged.
- **Shading:** A "shade net" or palm fronds *must* be placed over the nursery bed (about 1 meter high). This protects the tender seedlings from the harsh, direct sun and from the force of heavy rain.
- **Weeding:** Must be done regularly, by hand, as soon as weeds appear.
- **Thinning:** (From 3rd Term, Wk 8). Removing weak seedlings to give the strong ones space.

TRANSPLANTING

Transplanting is the process of moving the seedling from the nursery bed to the permanent field (which has already been ploughed and ridged).

- **HARDENING OFF (THE CRITICAL STEP):**

- This is the *most critical step* for survival. You cannot take a "soft" seedling from a protected, shaded, well-watered nursery and put it in a hot, sunny, harsh field; it will die from "**transplant shock**."
- **Process:** For 7-14 days *before* transplanting, you must "harden" (toughen) the seedlings.
- **How:**
 - i. **Gradually reduce watering** (let the soil dry out a bit more).
 - ii. **Gradually remove the shade cover** (expose them to more and more direct sun each day).
- This "toughens" the plant's cuticle and prepares it for the harsh conditions of the field.

- **Process of Transplanting:**

- Water the nursery bed *before* uprooting to make the soil soft.
- Uproot *carefully* (with a hand trowel) to keep as much soil *around the roots* as possible (the "root ball").
- Transplant *only* the healthy, strong, vigorous seedlings.
- **Timing:** Transplant in the **late afternoon or evening**, or on a cloudy day. This gives the plant the *entire cool night* to recover before the hot sun hits it.
- Water the seedling *immediately* after planting it in the field.

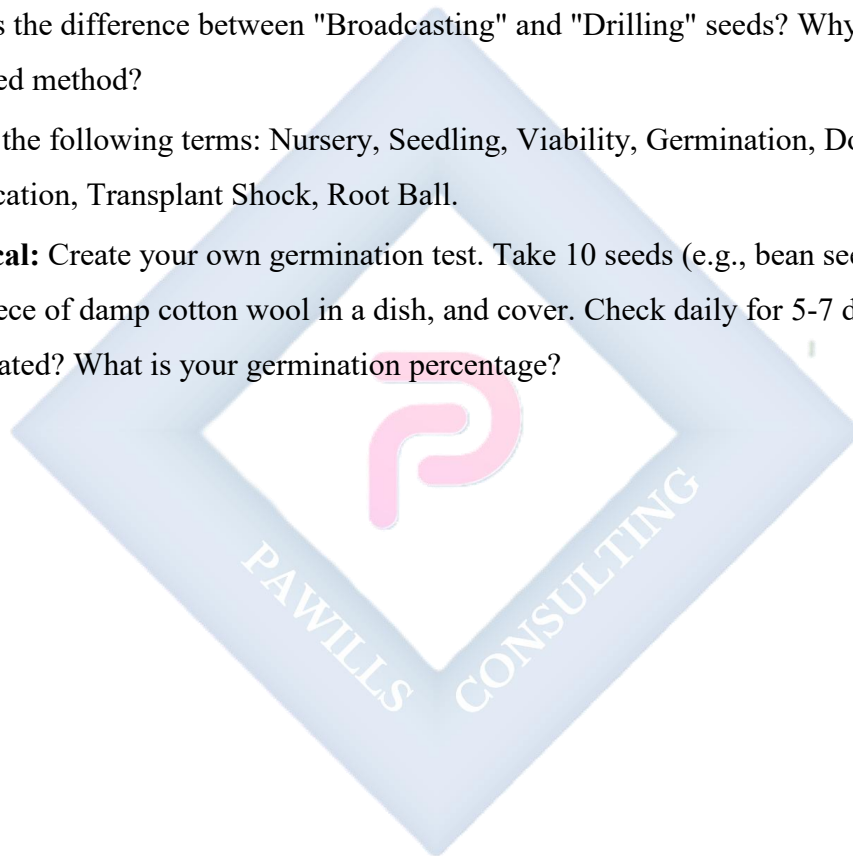
EVALUATION

1. List three main reasons *why* a farmer uses a nursery.
2. What is "Seed Viability," and how can you test for it?
3. What is "Seed Dormancy"? Describe the "Scarification" method used to break it.

4. What is the difference between a "Raised Bed" and a "Sunken Bed"?
5. What is "Hardening Off," and why is it the most critical step before transplanting?

ASSIGNMENT

1. Write a 250-word, step-by-step "Farmer's Guide" on how to successfully transplant a tomato seedling (from "Hardening Off" to "Watering in the field").
2. Draw a simple diagram of a "Raised Nursery Bed." Your diagram must include the bed itself, seedlings in "drills," and a "Shade Net" above it.
3. What is the difference between "Broadcasting" and "Drilling" seeds? Why is drilling the preferred method?
4. Define the following terms: Nursery, Seedling, Viability, Germination, Dormancy, Scarification, Transplant Shock, Root Ball.
5. **Practical:** Create your own germination test. Take 10 seeds (e.g., bean seeds), place them on a piece of damp cotton wool in a dish, and cover. Check daily for 5-7 days. How many germinated? What is your germination percentage?



WEEK 9: CROP PROPAGATION METHODS

✿ INTRODUCTION

Crop propagation is the process of *multiplying* or *creating new plants*. This is the fundamental basis of all crop production.

There are two "routes" a farmer can take to create a new plant: the sexual route (seeds) or the asexual route (cloning).



1. SEXUAL PROPAGATION (VIA SEEDS)

- **Definition:** This is creating a new plant from a **seed**.

- **Process:** It is "sexual" because it is the result of *pollination*—the fusion of male gametes (from the **pollen**) and female gametes (in the **ovule**). This fertilization creates an **embryo** (the seed), which is a *new, unique genetic individual*.
- **Advantages:**
 - a. **Easy and Cheap:** Seeds are generally easy to produce, buy, store, and transport (e.g., 1000 tomato plants can be held in one small seed packet).
 - b. **Creates New Varieties:** This is the *only* way to create new hybrid varieties. Crossing two different parents (Parent A + Parent B) creates genetic variation (a new combination).
 - c. **Virus-Free:** Many viruses that infect the parent plant *cannot* be transmitted through its seeds. It's a "clean" start.
- **Disadvantages (The "Big Problem"):**
 - a. **NOT "TRUE TO TYPE":** This is the key disadvantage. Because of genetic mixing (meiosis), a seed from a *very sweet* "Ogbomoso" mango will *not* grow into a tree that is identical to its parent. It will be a random mix, and the fruit might be sour or stringy.
 - b. **SLOW (Long Juvenile Period):** Many horticultural crops have a long "juvenile" or "childhood" period. A mango or orange tree grown from seed can take **7-10 years** before it produces its first fruit.

2. ASEXUAL (VEGETATIVE) PROPAGATION (VIA CLONING)

- **Definition:** This is creating a new plant from a *vegetative* part of the *parent* plant. This can be a piece of **stem, root, or leaf**. The new plant is a **CLONE**.
- **Result:** The new plant is **100% genetically identical** to its parent.
- **Advantages (The "Big Solution"):**
 - a. **"TRUE TO TYPE":** This is the main advantage. A cutting from a "Sweet Mango" tree will grow into a "Sweet Mango" tree. A cutting from a "Red Hibiscus" will *always* produce a "Red Hibiscus."

- b. **FAST:** It bypasses the "juvenile" period. A tree created by grafting can fruit in just **2-3 years** (instead of 10).
 - c. **Propagation of Seedless Fruit:** This is the *only* way to grow crops that do not produce viable seeds (e.g., modern **Bananas, Pineapples**).
- **Disadvantages:**
 - a. Can spread **viruses**. If the parent plant has a virus, the "clone" (cutting) will also have the virus.
 - b. More bulky and difficult to transport than seeds.

METHODS OF ASEXUAL (VEGETATIVE) PROPAGATION

- **CUTTINGS:**
 - **Definition:** A piece of a stem, root, or leaf is cut from the parent plant and planted in a nursery bed (or water), where it is induced to grow new roots.
 - **Examples:** This is the standard method for **Cassava** (stem cuttings), **Sugarcane** (stem cuttings), **Sweet Potato** (stem/vine cuttings), **Roses**, and **Bougainvillea**.
- **LAYERING:**
 - **Definition:** A stem is bent down and *induced* to grow roots *while still attached* to the parent plant.
 - **Process:** A branch is bent to the ground, a small wound is made on it, and it is covered with soil. *After* it grows its own roots, it is cut from the parent.
 - **Example:** Used for many climbing or vining plants that are difficult to root from a cutting.
- **SUCKERS, RUNNERS, SLIPS:**
 - **Suckers:** A new shoot that grows from the *base* or roots of the parent plant.
 - *Example:* This is the *main* way to plant **Bananas** and **Plantains** (using the "sword sucker").
 - **Runners (Stolons):** A horizontal stem that grows *above* the ground (e.g., strawberry plants).
 - **Slips:** A small plantlet that grows on the fruit.

- *Example:* The **crown** of a **Pineapple**.
- **GRAFTING AND BUDDING (The Most Advanced):**
 - **Definition:** This is the horticultural "surgery" of joining two different plants so they grow as one.
 - **The Parts:**
 - **Scion:** The *top* part (a bud or stem) from the *superior variety* (e.g., "Sweet Orange").
 - **Rootstock (Stock):** The *bottom* part (the root system) from a *hardy* variety (e.g., a "Sour Orange" root that is resistant to disease and drought).
 - **The Process:** The Scion is inserted into a carefully-made cut on the Rootstock and wrapped tightly. The "vascular systems" (the plant's "veins") grow together.
 - **Grafting:** Uses a *stem* (scion).
 - **Budding (T-budding):** Uses a single *bud* (scion).
 - **Use:** This is the **standard** for *all* commercial **Mangoes, Oranges, Avocados, Cashews, and Roses**.

EVALUATION

1. What is "Sexual Propagation"? What is its single biggest *disadvantage*?
2. What is "Asexual Propagation"? What is its single biggest *advantage*?
3. What does it mean for a plant to be "True to Type"?
4. Define "Grafting." What is the **Scion**, and what is the **Rootstock**?
5. What is the main method of propagating (a) Cassava, and (b) Bananas?

ASSIGNMENT

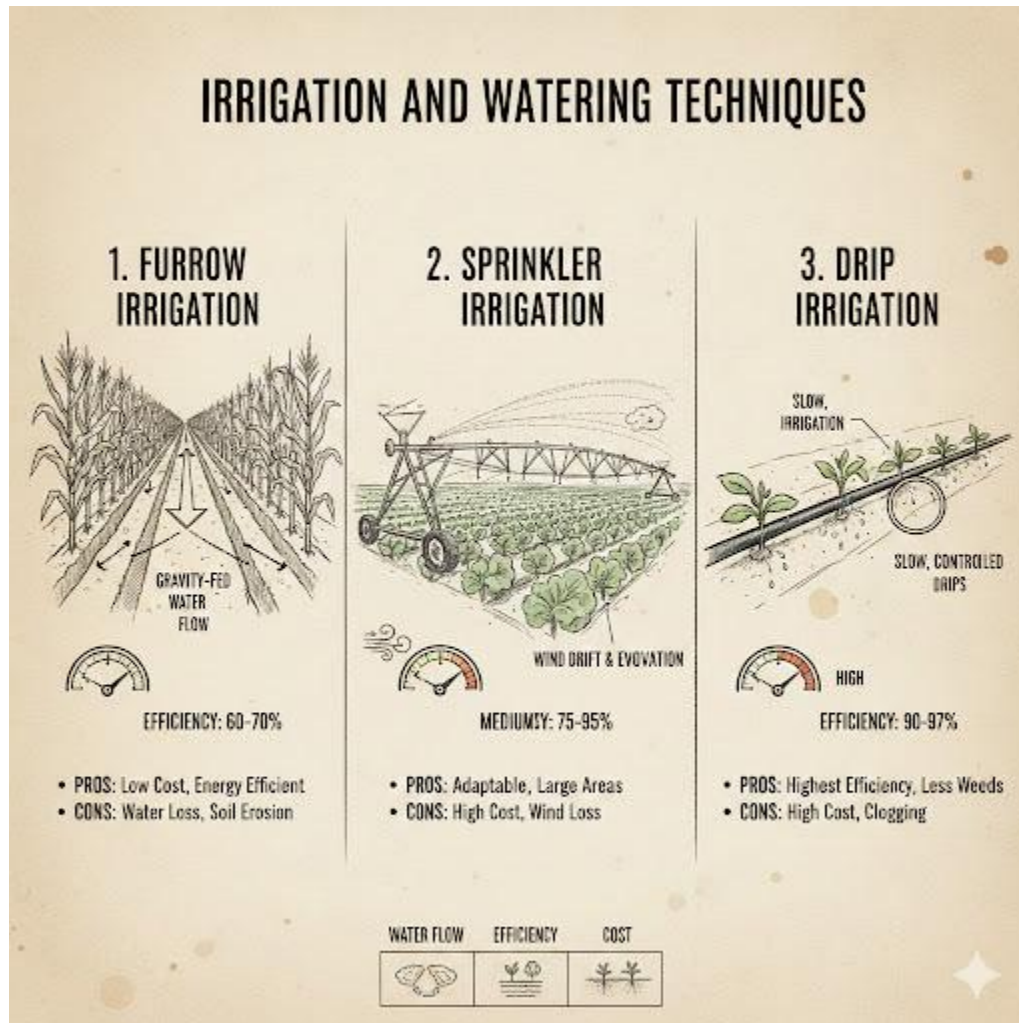
1. Write a 250-word essay on the topic: "If I want to start a commercial orange farm, why must I use grafted trees instead of seeds?" (Use the terms "true to type," "juvenile period," and "rootstock").
2. Draw a simple, labeled diagram of **T-Budding**. Show the rootstock, the T-cut, and the bud (scion) being inserted.

3. What is the *purpose* of using a hardy rootstock?
4. **Practical:** Propagate a plant asexually. (e.g., Take a stem cutting from a Sweet Potato vine and put it in water, or a Rose cutting, or plant the crown of a Pineapple). Write a one-paragraph report on what you did.
5. Define: Pollination, Clone, Juvenile Period, Cutting, Scion, Rootstock.



WEEK 10: IRRIGATION AND WATERING TECHNIQUES

◆ INTRODUCTION TO IRRIGATION



Irrigation is the **artificial application** of water to land or soil to supplement natural rainfall.

Why is it essential? In Nigeria, agriculture is "rain-fed," meaning it depends 100% on the rainy season (April-October). This has two massive problems:

1. Farming is impossible during the **dry season** (November-March).
2. Even in the rainy season, there can be a "dry spell" (e.g., "August Break") that can kill crops.

Irrigation is the technology that *breaks this limit*. It allows farmers (especially in the North) to grow high-value horticultural crops (like tomatoes and onions) **during the dry season**, when there is high sun, low disease, and **market prices are at their highest**.

METHODS OF IRRIGATION

1. SURFACE IRRIGATION

- **Definition:** Water is applied to the field and flows over the soil surface by **gravity**. This is the oldest, most common, and most "low-tech" method in the world.
- **Types:**
 - **Flood (or Basin) Irrigation:** The *entire* field is flooded, like a "basin" or "paddy."
 - *Use:* For crops that love water, like **Rice**, or in orchards.
 - **Furrow Irrigation:** This is the most common surface method for row crops. Water is run down small, parallel "channels" (furrows) that are dug *between* the crop ridges. The water soaks into the ridges from the side. *
- **Pros:**
 - **Very cheap** to set up. Requires minimal equipment (just pipes or a channel from a river/borehole).
- **Cons:**
 - **Extremely wasteful.** Up to 50% of the water is lost to **evaporation** (from the sun) and **runoff** (running off the field).
 - **Requires a level (flat) field.** It does not work on hills.
 - Can cause soil **erosion** (washing away topsoil).

2. SPRINKLER IRRIGATION

- **Definition:** Water is pumped through a network of pipes and sprayed *into the air* through "sprinkler heads." It falls on the crops like **artificial rain**.
- **Types:** Center-pivot (large, rolling machines for huge fields), solid-set (permanent pipes), portable sprinklers (like a lawn sprinkler).

- **Pros:**
 - Good for covering large, open areas (like lawns, maize fields, golf courses).
 - Can be used on *uneven* land (unlike furrow irrigation).
 - Cools the plants in very hot weather.
- **Cons:**
 - **Very wasteful.** A lot of water (15-30%) is lost to **evaporation** in the hot sun and **wind drift**.
 - **Promotes Fungal Diseases:** Keeping the leaves of crops (like tomatoes, squash) constantly *wet* is the #1 way to encourage the growth of **fungus** (like blight and mildew).
 - Requires high water pressure and expensive pumps.

3. DRIP (OR TRICKLE) IRRIGATION

- **Definition:** This is the **most advanced** and **most efficient** modern method. It involves a network of plastic pipes (drip lines) that are laid on or near the soil. Water is delivered *very slowly* (one drop at a time) through small "emitters."
- **Key Principle:** Water is applied *directly to the root zone* of *each individual plant*. You water the *plant*, not the *soil*.
- **Pros:**
 - a. **Extremely Efficient:** 90-95% of the water is used by the plant. *No* water is wasted on weeds, paths, or evaporation.
 - b. **Reduces Disease:** The plant's leaves stay **100% dry**, which drastically reduces the risk of **fungal growth**.
 - c. **Fertigation:** This is a major advantage. The farmer can *dissolve* fertilizer in the irrigation water and deliver nutrients *directly to the roots* in small, precise doses ("feeding" the plant every day).
 - d. Works with low pressure and on any terrain.
- **Cons:**
 - **High Initial Cost:** Expensive to buy the pipes, filters, and emitters.

- **Clogging:** The tiny emitters can be easily *clogged* by dirty water or sand (requires good filters).
- **Use:** This is the **best** method for high-value, row-crop horticulture (tomatoes, peppers, orchards, greenhouses).

EVALUATION

1. What is "irrigation," and why is it a "game-changer" for dry-season farming?
2. Describe **Furrow Irrigation**. What is its main *advantage* and main *disadvantage*?
3. What is the *biggest disadvantage* of Sprinkler Irrigation, especially for tomatoes? (Hint: Disease).
4. What is **Drip Irrigation**? Explain its "key principle."
5. What is "**Fertigation**," and which irrigation method makes it possible?

ASSIGNMENT

1. Write a 250-word essay arguing which irrigation system is the *best* for a small-scale, commercial tomato farmer in Northern Nigeria. (You must compare Drip vs. Furrow).
2. Draw a simple diagram for each of the three main irrigation methods (Furrow, Sprinkler, Drip).
3. What is a "rain-fed" farm? What is its main limitation?
4. Why is Drip Irrigation so efficient? (Where does it save water?)
5. Define: Evaporation, Runoff, Waterlogging, Emitter, Fertigation.

WEEK 11: REVISION

GENERAL REVIEW OF ALL FIRST TERM TOPICS

This week is for a comprehensive review of all topics covered this term. We have built a complete foundation in horticulture, from definition to soil, to propagation, to watering.

- **WEEKS 1-3 (THE "WHAT"):** We defined **Horticulture** (intensive "garden" cultivation) and contrasted it with **Agronomy** (extensive "field" cultivation). We learned its **Importance** (Nutrition, Economy, Aesthetics). We identified its **Branches** (Olericulture, Pomology, Floriculture, Landscaping) and **Classifications** (Food, Cash, Ornamental, Medicinal).
 - *Key Concept:* A single crop (like Ginger) can be a *food crop*, *cash crop*, and *medicinal crop* all at once.
- **WEEK 4 (THE "TOOLS"):** We identified, learned the use of, and learned the care for all essential tools.
 - *Key Concepts:* **Spade** (digs) vs. **Shovel** (moves). **Secateurs** (clean cuts for pruning) vs. **Cutlass** (heavy clearing). **Knapsack Sprayer** (chemicals, *must* be washed). **Watering Can** (must have a "**rose**" for seedlings).
- **WEEK 6 (THE "FOUNDATION"):** We learned about **Soil**.
 - *Key Concepts:* **Sand** (drains well, low nutrients) vs. **Clay** (holds water, poor drainage) vs. **Loam** (ideal).
 - *Soil Management:*
 - **Manure/Compost:** Fixes *both* sandy and clay soils.
 - **Liming:** *Raises* the pH of *acidic* soil (unlocks nutrients).
 - **Mulching:** Covers soil to *save water* and *stop weeds*.
 - **Crop Rotation:** Prevents disease/pests and manages nutrients.
- **WEEK 8 (THE "START"):** We learned about **Nursery Practices**.
 - *Key Concepts:* Nurseries *protect* vulnerable seedlings (e.g., tomato, pepper).
 - *Dormancy:* Breaking the seed coat (**Scarification**).

- *Management*: Use **drills** (lines) for sowing, **shade** (palm fronds) for protection, and **watering can with rose** (gentle).
- *Transplanting*: Must first perform "**Hardening Off**" (gradually reducing water/shade) to prevent "**transplant shock**." Best done in the *evening*.
- **WEEK 9 (THE "GENERATION")**: We learned **Propagation**.
 - **Sexual (Seed)**: Easy, cheap, but **NOT "true to type"** and **slow** (long juvenile period).
 - **Asexual (Vegetative)**: A *clone*. "**True to type**" and **fast** (fruits in 2-3 years).
 - *Methods*: **Cuttings** (Cassava), **Suckers** (Banana), and **Grafting/Budding** (Mango, Orange).
 - *Grafting*: **Scion** (good fruit) + **Rootstock** (hardy root).
- **WEEK 10 (THE "WATER")**: We learned **Irrigation**.
 - *Why?* To farm in the **dry season**.
 - *Surface (Furrow)*: Cheap, low-tech, but *very wasteful*.
 - *Sprinkler*: Like rain, but wastes water to evaporation and *promotes fungal disease*.
 - *Drip*: **Most efficient**. Saves water, reduces disease, allows for "**fertigation**" (applying fertilizer with the water).

EVALUATION

1. What is the main difference between Pomology and Olericulture?
2. What is the single most important soil management practice, and how does it help *both* sandy and clay soils?
3. What is "Hardening Off," and why is it essential?
4. What are the two main *advantages* of Asexual (Vegetative) propagation over Sexual (Seed) propagation?
5. What is "Fertigation," and which irrigation method makes it possible?

ASSIGNMENT

1. **Project:** You are a small farmer in Nigeria. You want to grow one hectare of high-quality **tomatoes** for sale in the *dry season*. Write a one-page "Production Plan" that addresses:
 - (a) How you will propagate the plants (Week 9).
 - (b) How you will raise the young plants (Week 8).
 - (c) How you will manage the soil's pH and fertility (Week 6).
 - (d) How you will water your crop (Week 10).
 - (e) The tools you will need (Week 4).
2. Draw a simple diagram of "Grafting," showing and labeling the **Scion** and the **Rootstock**.
3. Create a table comparing the 3 irrigation methods (Surface, Sprinkler, Drip) based on **Cost, Water Efficiency, and Disease Risk**.
4. Define the 10 most important terms you learned this term.
5. What is the difference between a "Cash Crop" and a "Food Crop"? Give an example of a crop that is both.



SS1 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

SECOND TERM

WEEK 1: FERTILIZER AND MANURE APPLICATION

🌿 INTRODUCTION TO SOIL FERTILITY AND NUTRIENTS

FERTILIZER AND MANURE APPLICATION		
Feature	Organic Fertilizers	Inorganic Fertilizers
Source	Plant/Animal waste (Manure, Compost)	Chemical synthesis (N-P-K blends)
	Slow, gradual	Fast, immediate
Nutrient Release	Slow, gradual	
Effect on Soil	Improves soil structure, microbes	Provides nutrients only, can harm soil
Cost	Generally cheaper (DIY)	Generally more expensive (Industrial)
Environmental Impact	✓ Eco-friendly, reduces waste	✗ ✗
	✓ Eco-friendly, reduces waste	✓ Pollution risk (leaching)
Examples	Compost, Manure, Bone Meal	Urea, DAP, Ammonium Nitrate

Plants, like all living things, need to "eat" to grow. Their food is a set of chemical elements (nutrients) which they absorb from the soil through their roots. When we harvest a crop (like a tomato or a leaf of *ugu*), we are removing those nutrients from the soil.

If we do this repeatedly, the soil becomes "hungry," infertile, and "exhausted," leading to poor growth and low yields. **Fertilizers** and **manures** are substances (natural or synthetic) that are

added to the soil to *replenish* these lost nutrients, ensuring that the next crop can grow strong and healthy.

Macro-Nutrients: Plants need three main nutrients in large amounts. These are known as **NPK**:

- **N = Nitrogen:** The "Leaf Maker." Essential for vegetative growth (leaf and stem production) and gives plants their dark green color (it's a key part of chlorophyll).
- **P = Phosphorus:** The "Root and Fruit Maker." Essential for strong root development, and also for flowering and fruiting.
- **K = Potassium:** The "Plant Health Guard." Essential for overall plant vigor, disease resistance, water regulation, and fruit quality.

ORGANIC FERTILIZERS (MANURES)

Definition: Organic fertilizers are derived from the remains, by-products, or waste products of **living organisms** (plants and animals). They are natural materials that "feed the soil, which feeds the plant."

Types of Organic Manures:

1. **Farm Yard Manure (FYM):** This is a mixture of animal waste (dung/feces), urine, and the "bedding" material (like straw or wood shavings) used in animal houses. (e.g., cattle, goat, sheep manure). It must be **composted (rotted)** before use.
2. **Poultry Manure:** This is the waste from poultry (chickens, turkeys). It is extremely rich and "**hot**" (very high in Nitrogen) and *must* be composted before use, or it will "burn" and kill young plants.
3. **Compost:** This is the dark, crumbly, nutrient-rich material (humus) produced when farmers *intentionally* pile up and decompose organic materials like kitchen scraps, grass clippings, leaves, and weeds.
4. **Green Manure:** This is a "living" manure. A farmer plants a specific crop (usually a legume, like beans, *Mucuna*, or *Centrosema*) and then *plows it into the soil* while it is still green. The decomposing plants release a large amount of nutrients (especially nitrogen) for the *next* crop.

Advantages of Organic Manure:

- **Improves Soil Structure:** This is its **most important benefit**. It adds **humus** (organic matter), which acts like a "sponge."
 - In **clay soil**, it separates the sticky particles, making it looser and improving drainage.
 - In **sandy soil**, it fills the large gaps, helping it *hold* water and nutrients.
- **Slow-Release of Nutrients:** Nutrients are released slowly as the manure is decomposed by soil microbes, feeding the plants over a long period.
- **Feeds Soil Life:** It provides food for essential soil organisms (like earthworms, fungi, and bacteria), which are vital for a healthy soil ecosystem.
- **Low Cost:** Can often be sourced cheaply or for free on the farm.
- **Water Retention:** Increases the soil's ability to hold water.

Disadvantages of Organic Manure:

- **Bulky and Difficult to Handle:** You need to transport and spread *a lot* of it (tons per hectare) to get enough nutrients.
- **Slow-Acting:** Nutrients are not immediately available to the plants.
- **Variable Composition:** The exact NPK value is unknown and can vary.
- **May Contain Pests/Weeds:** If not properly composted, manure can introduce harmful weed seeds and pathogens (diseases) into the soil.

INORGANIC FERTILIZERS (CHEMICAL/SYNTHETIC)

Definition: Inorganic fertilizers (also called "chemical" or "synthetic") are mineral salts manufactured in a factory. They are "artificial" fertilizers designed to be fast-acting.

Types of Inorganic Fertilizers:

1. **Straight Fertilizers:** These supply only *one* of the major nutrients (N, P, or K).
 - **Urea:** Supplies Nitrogen (N).
 - **Single Superphosphate (SSP):** Supplies Phosphorus (P).

- **Muriate of Potash (MOP):** Supplies Potassium (K).
- 2. **Mixed/Compound Fertilizers (NPK):** These supply two or more nutrients combined in one granule.
 - **Example: NPK 15:15:15.** This is the most common. The numbers mean the bag contains 15% Nitrogen, 15% Phosphorus, and 15% Potassium.
 - **Example: NPK 20:10:10.** This is high in Nitrogen (for leafy greens or maize).

Advantages of Inorganic Fertilizers:

- **Fast-Acting:** The nutrients are water-soluble and in a form the plant can use *immediately*.
- **Concentrated:** The nutrient content is very high, so you only need to apply a small, non-bulky amount.
- **Specific:** You know *exactly* what nutrients you are applying and in what amount (e.g., NPK 15:15:15).
- **Easy to Handle/Store:** They are usually sold in clean, 50kg bags.

Disadvantages of Inorganic Fertilizers:

- **Expensive:** They represent a high cash cost for the farmer.
- **Leaching:** Because they are so soluble, they can be easily *washed away* by heavy rain. This wastes money and pollutes groundwater and rivers (causing "eutrophication").
- **Damages Soil Structure:** They do *not* add organic matter. Overuse can make soil hard, compact, acidic, and kill off the beneficial soil microbes.
- **"Burning":** If applied in excess or too close to the plant's stem or roots, the high salt concentration will "burn" (dehydrate) and kill the plant.

METHODS OF APPLICATION

How you apply the fertilizer is just as important as the fertilizer itself. The goal is to get the nutrient to the roots *without* wasting it.

1. Broadcasting:

- **Method:** Spreading the fertilizer (by hand or machine) *evenly* over the *entire* surface of the field.
- **Use:** For crops that are planted close together (like lawns, wheat, or spinach) or *before* planting and ploughing. It is fast but can be wasteful.

2. Ring Method:

- **Method:** Applying the fertilizer in a *circular band* around the base of a plant, but *away* from the main stem.
- **Use:** This is the best method for **trees** (e.g., orange, mango, cocoa) and large, individual plants (like eggplant or garden egg).
- **Why?** The fertilizer is placed under the "**drip line**" (the edge of the canopy), which is where the active "feeder roots" are.

3. Side/Band Placement:

- **Method:** Applying the fertilizer in a *line* or "band" *beside* the row of crops (usually 5-8cm away from the plants).
- **Use:** This is the most efficient method for **row crops** (like maize, vegetables, okra) as it places the nutrients *close to the roots* where they are needed, without wasting it in the paths between rows.

4. Foliar Application:

- **Method:** Dissolving a special liquid fertilizer in water and using a **knapsack sprayer** to spray it *directly onto the leaves* of the plant.
- **Use:** This is a "quick fix" or *emergency* method used to correct a specific nutrient deficiency (like a lack of iron or calcium) very quickly, as the leaves absorb it instantly. It is *not* a replacement for soil fertilization.

EVALUATION

1. What are the three "macro-nutrients" represented by **NPK**, and what is the primary function of each in a plant?
2. What is the *most important advantage* of organic manure that is *not* related to nutrients? (Hint: Soil...).

3. List three advantages and three disadvantages of inorganic (chemical) fertilizers.
4. What is the difference between a "Straight" and a "Compound" fertilizer? What does "NPK 20:10:10" mean?
5. Describe the "**Ring Method**" of application. For what type of crop is it best suited, and why?

ASSIGNMENT

1. Write a 250-word essay comparing and contrasting organic manure and inorganic fertilizers. Which would you recommend for long-term soil health, and why?
2. What is "**Green Manure**"? Research and name one plant that is commonly used as green manure in West Africa.
3. A farmer applies Urea (a nitrogen fertilizer) too close to the stem of his young tomato plants, and they wilt and die. Explain what happened using the term "**burning**."
4. Draw simple diagrams illustrating the **Broadcasting, Ring, and Side Placement** methods of fertilizer application.
5. Define the following terms: Humus, Leaching, Eutrophication, Drip Line, Foliar Application.

WEEK 2: WEED AND PEST CONTROL

❁ INTRODUCTION TO WEEDS

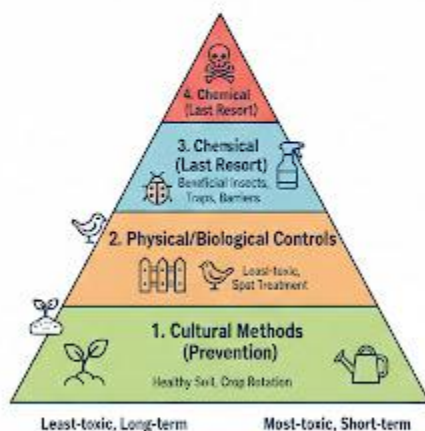
A **weed** is any plant growing where it is not wanted. It is a "plant out of place."

- An *Ugu* plant growing in a tomato farm is a weed.
- A maize plant growing in a rice paddy is a weed.
- A *Speargrass* or *Waterleaf* plant growing anywhere in a farm is a weed.

Weeds are the single biggest cause of yield loss in many farms.

Integrated Pest Management (IPM)

The IPM Pyramid



COMMON PESTS BY DAMAGE



Chewing Mouthparts
Caterpillars, Beetles, Beetles, Slugs



Sucking Mouthparts
Aphids, Whiteflies, Mites

HARMFUL EFFECTS OF WEEDS

Weeds are harmful because they are aggressive and out-compete the desired crop.

1. **COMPETITION (The Main Effect):** This is the biggest problem. Weeds are "thieves" that steal the resources you provided for your crops. They compete for:
 - **Water:** Weeds often have very deep or very dense root systems that "steal" water from the crop's root zone.
 - **Nutrients:** Weeds absorb the expensive fertilizer you applied for your crop, leading to nutrient deficiency in the crop.
 - **Light:** Tall weeds (like elephant grass or *Chromolaena*) grow faster than crops and form a "canopy," blocking sunlight and "shading out" the smaller crop plants (like peppers), which then die.
 - **Space:** They "crowd out" the crop, reducing airflow.
2. **HOSTING PESTS AND DISEASES:** Weeds can act as an alternative "home" (host) for insects and pathogens. Pests (like aphids) can live on the weeds and then move onto your crop.
3. **REDUCED HARVEST QUALITY:** Weed seeds (like *Striga* or *Tridax*) can mix with harvested grain (like rice), reducing its purity, quality, and market value.
4. **PHYSICAL OBSTRUCTION:** Thorny weeds (like "touch-and-die" *Mimosa*) make the farm difficult and painful to walk through and harvest.

METHODS OF WEED CONTROL

A. CULTURAL METHODS

(Using good *farming practices* to prevent weeds)

1. **Mulching:** (From 1st Term). Covering the soil with a layer (e.g., straw, grass clippings, black plastic) **blocks sunlight**, preventing weed seeds from germinating. This is the *most effective* cultural method.

2. **Proper Spacing:** Planting crops at the correct density (not too wide) allows them to quickly grow and form a "canopy" (a roof of leaves) that shades the ground, suppressing weeds.
3. **Cover Cropping:** Planting "living mulches" (like melon, sweet potato vines, or *Mucuna*) *between* crop rows to "choke out" and smother weeds.

B. PHYSICAL / MECHANICAL METHODS

(Using tools or hands)

1. **Hand-Pulling:** Physically pulling the weeds out by hand. Effective for small gardens or in nursery beds, but slow and very labor-intensive.
2. **Hoeing / Slashing:** Using a **hoe** or **cutlass** to cut the weeds at or below the soil level. This is the most common method in Nigeria.
3. **Mowing / Slashing:** Using a lawnmower or slasher to cut weeds (e.g., in an orchard between the trees).

C. BIOLOGICAL METHODS

(Using living organisms to control weeds)

- **Method:** Releasing a specific *insect* or *pathogen* that *only* eats the target weed and will not harm the crop. (This is advanced and less common).
- **Simple Example:** Using **goats or sheep** to graze in a tree crop (e.g., oil palm or mango) orchard. They eat the weeds but can't reach the high fruit.

D. CHEMICAL METHODS

(Using chemicals (herbicides) to kill weeds)

- **Herbicides:** A **herbicide** is a chemical that kills plants.
- **Types of Herbicides:**
 - a. **Selective Herbicides:** This is the "smart" type. It is designed to kill *only* a specific *type* of plant.

- *Example:* A "broadleaf" herbicide (like 2,4-D) will kill *only* broadleaf weeds (like *Chromolaena*, *Tridax*) but is **safe** for grass crops (like **Maize, Rice, Lawns**).
- b. **Non-Selective Herbicides:** (e.g., **Glyphosate** / "Roundup," Paraquat). This is the "total" killer. It **kills every plant it touches** (both weeds *and* your crop).
 - *Use:* It is used for clearing a field *before* planting, or for spraying *very carefully* in the paths between rows.
- c. **Pre-Emergence:** A herbicide applied to the *soil before* the crop or weeds have sprouted. It kills the weeds as they germinate.
- d. **Post-Emergence:** A herbicide applied *onto the leaves* of weeds that are *actively growing*.

2. PESTS

Definition: A **pest** is any animal or organism that is harmful to humans, their animals, or their crops.

- **Insect Pests:** (The most common)
- **Vertebrate Pests:** Rodents (rats, grasscutters), Birds (which eat seeds and fruits).
- **Other:** Nematodes (microscopic soil worms - *see Week 3*), Mites, Snails.

Types of Insect Pests (Classified by Mouthpart/Damage):

1. BITING AND CHEWING PESTS:

- **Examples:** Grasshoppers, Caterpillars (Larvae of butterflies/moths), Beetles (e.g., Yam Beetle), Termites.
- **Damage:** They have jaws (mandibles) and **eat holes** in the leaves, stems, roots, and fruits. They reduce the surface area for photosynthesis.

2. PIERCING AND SUCKING PESTS:

- **Examples:** Aphids, Mealybugs, Whiteflies.
- **Damage:** They have a needle-like mouthpart (proboscis) that they use to *pierce* the plant and *suck* the **sap** (the plant's "blood").

- **Secondary Damage (CRITICAL):**

- i. They are the main **vectors** (transmitters) for **plant viruses** (like Leaf Curl Virus).
- ii. They excrete a sticky, sugary "honeydew" that leads to a *black, sooty mold* growing on the leaves.

METHODS OF PEST CONTROL

(This is an introduction to IPM - Integrated Pest Management)

- **Cultural:** **Crop rotation** (starves pests), **sanitation** (removing crop debris to destroy eggs), **companion planting** (planting marigolds, which repel nematodes), **timing** (planting early to avoid pests).
- **Physical:** **Hand-picking** (e.g., removing large caterpillars), **traps** (e.g., yellow sticky traps for whiteflies), **nets** (to keep birds off fruit).
- **Biological:** Using **natural predators** to control pests (e.g., releasing **Ladybugs** to eat aphids, or using **Neem oil** spray).
- **Chemical:** Using **Pesticides** (a general term).
 - **Insecticide:** Kills insects.
 - **Rodenticide:** Kills rodents (rat poison).
 - **Nematicide:** Kills nematodes.
 - **Avicide:** Kills birds.

EVALUATION

1. What is a weed? List the 4 key things that weeds compete with crops for.
2. What is the difference between a **Selective** and a **Non-Selective** herbicide?
3. What are the two main types of insect pests (classified by their mouthparts)?
4. Give an example of (a) a cultural, (b) a physical, and (c) a chemical method of *weed* control.
5. What is a "vector" in the context of plant pests, and which pest type is the main vector?

ASSIGNMENT

1. Write a 250-word report on **Mulching**. Explain how this single cultural practice can control weeds, conserve water, and improve the soil, all at the same time.
2. Research "Companion Planting." Name two plants that are "companions" (help each other when planted together) and two that are "antagonists" (hurt each other).
3. What is "Integrated Pest Management (IPM)"? (This is the modern approach that *combines* cultural, biological, and chemical methods).
4. Find a common weed in your school compound or house. Draw it, and describe why it is a successful competitor (e.g., "it grows very fast," "it has deep roots," "it produces many seeds").
5. Define: Herbicide, Pesticide, Vector, Crop Rotation, Canopy, Leaching, Pre-emergence, Post-emergence, Selective, Non-selective.



WEEK 3: CROP DISEASES

INTRODUCTION

Definition: A **plant disease** is any abnormal condition in a plant that interferes with its normal structure, function, or economic value. It is a "**deviation from the normal state of health**" and is caused by a persistent agent called a **pathogen**.

It is crucial to distinguish disease from pest damage:

- **Pest Damage:** Caused by an *insect* (e.g., a hole in a leaf). You can often see the pest.
- **Disease:** Caused by a *microscopic pathogen* (e.g., a spot on a leaf, a wilt).

CROP DISEASES: PATHOGEN TYPES			
CAUSES, SYMPTOMS, & CONTROL			
PATHOGEN TYPE	CAUSE	COMMON SYMPTOMS	CONTROL METHODS
 Fungi		Leaf spots, mold, rusts, blights, blights, root rot	Fungicides, crop rotation, resistant varieties
Bacteria	Single-celled microorganisms	Bacterial soft rots, leaf spots, galls	Antibiotics (limitation), copper-based sprays
	Infectious agents, require host	Stunted growth, mosaic patterns, leaf curl, deformities	Remove infected plants, (insect, planting material)
 Nematodes	Microscopic roundworms	Root knots, stunted growth, wilting, nutrient deficiency symptoms	Nematicides, soil solarization, cover crops

Effective crop disease management relies on accurate identification and integrated strategies.

1. CAUSATIVE AGENTS (PATHOGENS)

It is crucial to know *what* is causing the disease, as the control method is completely different for each.

1. FUNGI (Singular: Fungus):

- **This is the MOST COMMON cause of plant disease.**
- **Description:** Fungi are microscopic organisms (like mold or mushrooms) that are *not* plants (they cannot photosynthesize). They are parasites that "eat" the plant.
- **Spread:** They spread via tiny, dust-like **spores** (which are like seeds) that travel in the wind, water (rain-splash), and on dirty tools.
- **Favorable Conditions:** They *thrive* in **high humidity, high moisture, and poor air circulation**. (This is why they are a huge problem in the rainy season in Southern Nigeria).
- **Examples of Diseases:**
 - **Damping-Off:** Kills new seedlings right at the soil level. The stem rots and the seedling falls over.
 - **Leaf Spots:** (e.g., Black Spot on roses, Cercospora leaf spot on *ugu*).
 - **Powdery Mildew:** A white, "powdery" growth on the surface of leaves.
 - **Root Rot / Stem Rot:** The plant rots from the base or roots, often caused by waterlogged soil.
 - **Blight:** (e.g., *Late Blight* of tomato/potato, a very fast-killing, water-soaked lesion).
- **Control: Fungicides** (chemicals that kill fungi).

2. BACTERIA (Singular: Bacterium):

- **Description:** Microscopic, single-celled organisms.
- **Spread:** Bacteria *cannot* enter a plant's waxy cuticle on their own. They must enter through a **wound** (e.g., from insect damage, pruning, or hail) or a natural opening (like stomata). They spread via water-splash.
- **Examples of Diseases:**

- **Bacterial Blight:** (e.g., on cassava, beans). Causes large, irregular, "water-soaked" spots on leaves.
- **Soft Rot:** Causes vegetables (like carrots, cabbage) to turn into a slimy, foul-smelling, mushy mess.
- **Bacterial Wilt:** Clogs the plant's "veins" (xylem), causing the entire plant to wilt and die suddenly.
- **Control: Bactericides** (less common) or by controlling insect vectors and sanitation.

3. VIRUSES:

- **Description:** Sub-microscopic particles (much smaller than bacteria). They are not truly "alive." They are bits of DNA/RNA that *hijack* the plant's cells to reproduce.
- **Spread (CRITICAL):** Viruses are *almost always* transmitted by **insect vectors** (like **aphids, whiteflies, thrips**) that feed on an infected plant and carry the virus to a healthy one (like a "dirty needle"). They can also be spread by *dirty tools* (like secateurs).
- **Examples of Diseases:**
 - **Cassava Mosaic Disease (CMD):** Causes a yellow-green "mosaic" pattern and deformed, twisted leaves.
 - **Tomato Yellow Leaf Curl Virus (TYLCV):** Stunts the plant, makes the new leaves small, yellow, and "curled" up.
- **Control: There is NO CURE for a plant virus.**
 - **Control Method:** You must (1) **control the insect vector** (spray insecticides) and (2) **remove (rogue) and burn** all infected plants to stop the spread.

4. NEMATODES:

- **Description:** Microscopic, parasitic **roundworms** that live in the soil. They are animals, not microbes.

- **Damage:** They attack the **roots** of the plant, using a sharp "stylet" to pierce and suck the root cells.
- **Example: Root-Knot Nematodes.** This is the most famous. They cause the roots to form ugly, abnormal *swellings* ("galls" or "knots").
- **Symptoms:** Because the roots are damaged, the plant *above ground* looks stunted, yellow, and wilts easily in the hot sun, even if the soil is moist.
- **Control: Nematicides** (strong soil chemicals) or **Crop Rotation** with a "non-host" crop (like marigolds, which repel them).

2. SYMPTOMS (IDENTIFICATION OF DISEASE)

- **WILT:** The plant is drooping, as if it has no water (even if the soil is wet). Caused by root rot (fungi) or bacterial wilt.
- **LEAF SPOTS:** Discolored (brown, black, yellow) spots on the leaves.
- **BLIGHT:** A rapid, widespread "burning" or dying of leaves and stems.
- **MOSAIC:** A pattern of light-green or yellow "mosaic" shapes on the leaf. (A classic virus symptom).
- **GALLS (KNOTS):** Abnormal swellings or "tumors" on the stems or (for nematodes) on the roots.
- **DAMPING-OFF:** Young seedlings suddenly fall over as their stem rots at the soil line (a fungal disease).
- **ROTTING:** Soft, mushy, foul-smelling decay of fruits or roots (bacterial or fungal).

3. PREVENTION AND CONTROL METHODS

Prevention is *always* better, cheaper, and more effective than cure.

A. PREVENTION METHODS (PROPHYLACTIC)

1. **Use Resistant Varieties:** This is the **#1 best method**. Plant breeders have developed many varieties that are naturally immune or resistant to common diseases (e.g., CMD-resistant cassava, TYLCV-resistant tomatoes).

2. **Crop Rotation:** Do not plant the same crop (or its relatives) in the same soil for 3-4 years. This *starves* the soil-borne pathogens (like fungi and nematodes) that are waiting for that specific crop.
3. **Sanitation (Farm Hygiene):**
 - **Roguing:** Uproot and *burn* (do not compost) any plant that shows signs of disease (especially viruses).
 - **Tool Cleaning:** Disinfect tools (like secateurs, knives) with bleach or alcohol when moving between plants.
 - **Remove Debris:** After harvest, clear all old plant debris from the field, as pathogens can survive on it.
4. **Proper Spacing:** Good airflow between plants ensures leaves dry quickly, which **prevents fungal growth.**
5. **Proper Watering:** Water the *soil* (e.g., with drip irrigation), not the *leaves*. Wet leaves are a "welcome mat" for fungi.

B. CONTROL METHODS (CURATIVE)

(Used *after* a disease has already appeared)

1. **Chemical Control:** Applying a chemical to kill the pathogen.
 - **Fungicides:** To control fungi (e.t., Copper-based sprays like Bordeaux Mixture, or synthetics like Mancozeb).
 - **Bactericides:** To control bacteria (less common, copper can work).
 - **Nematicides:** To control nematodes (very toxic, handle with care).
 - **Insecticides:** To control the *vectors* that spread viruses.
2. **Physical Control:**
 - **Roguing:** Removing and destroying infected plants to stop the spread.

EVALUATION

1. What is a "pathogen"? List the 4 main types of pathogens that cause crop diseases.

2. What is the *most common* group of pathogens, and what environmental condition do they love?
3. A farmer's tomatoes have yellow, curled leaves and are stunted. What *type* of pathogen is the likely cause, and how is it probably being spread?
4. What is the *only* way to control a plant virus, and why?
5. What is "Crop Rotation," and how does it help control *both* diseases (Week 3) and pests (Week 2)?

ASSIGNMENT

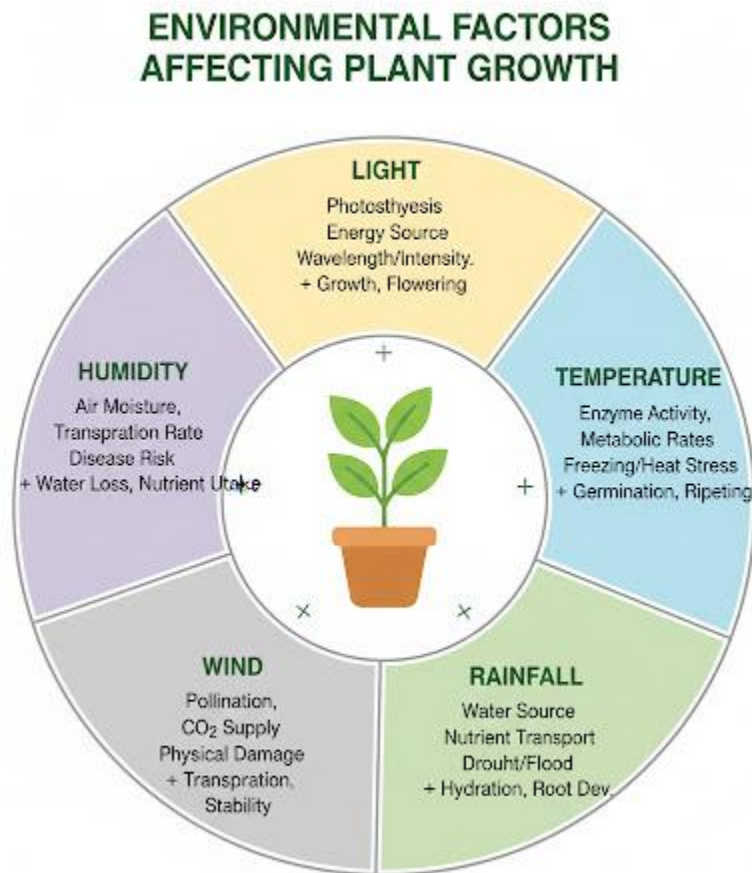
1. Write a 250-word report on **Cassava Mosaic Disease (CMD)**. What is its cause, its vector (transmitter), its symptoms, and the *best* way for a farmer to prevent it?
2. What is "**Damping-Off**"? Why is it a major problem for a nursery (from 1st Term)?
3. Go outside and inspect 5 different plants (e.g., mango, hibiscus, grass, or any "weed"). Find a leaf that is not perfectly healthy. Draw the "symptom" you see (e.g., a spot, a hole, a mosaic) and (if you can) guess what *type* of agent (insect, fungus, virus) might have caused it.
4. What is a **Resistant Variety**? Why is this the *best* method of disease control for a farmer?
5. Define: Fungicide, Vector, Pathogen, Roguing, Waterlogging, Nematode, Mosaic, Galls.

WEEK 4: ENVIRONMENTAL FACTORS

☀ INTRODUCTION

These are the **abiotic** (non-living) climatic and soil factors that govern all life on the farm. A farmer cannot *control* these factors (like they can pests), but they must *understand* them, *select crops* based on them, and *manage* their effects.

These factors (also called **ecological factors**) determine *what* can be grown, *where* it can be grown, and *when* it can be grown.



1. LIGHT (SOLAR RADIATION)

- **IMPORTANCE:** This is the most fundamental factor. Light is the **energy source** for **photosynthesis**.
 - **Photosynthesis:** The process where plants use light energy, water, and carbon dioxide (CO_2) to create sugar (their food/energy) and release oxygen (O_2).
- **KEY ASPECTS:**
 - a. **Light Intensity (Brightness):**
 - **High Intensity:** Leads to high rates of photosynthesis and fast growth. Most field crops are "sun-loving."
 - *Examples:* **Maize, Sugarcane, Tomatoes, Peppers, Yams.**
 - **Low Intensity (Shade):** Some horticultural crops *prefer* or *require* shade. Direct, harsh sun will "scorch" their leaves.
 - *Examples:* **Cocoa, Coffee** (are traditionally grown *under* larger "shade trees"). Many indoor ornamental plants (like ferns).
 - b. **Light Duration (Photoperiod):**
 - **Definition:** The length of the day (the number of hours of light in a 24-hour period).
 - **Importance:** This is the *biological trigger* for **flowering** in many plants.
 - **Short-Day Plants:** Will *only* flower when the days are *short* (e.g., less than 12 hours of light).
 - *Examples:* **Hibiscus/Zobo, Soybean, Chrysanthemum**, many traditional rice varieties.
 - **Long-Day Plants:** Will *only* flower when the days are *long* (e.g., more than 12 hours). This is why some temperate crops *cannot* produce seed in Nigeria (our days are too short).
 - *Examples:* **Cabbage, Lettuce, Carrots, Wheat.**
 - **Day-Neutral Plants:** Flower based on *age* (maturity), not day length.
 - *Examples:* **Tomato, Maize**, most modern hybrid crops.

2. TEMPERATURE (HEAT)

- **IMPORTANCE:** Temperature controls the *rate* of all chemical reactions in the plant (photosynthesis, respiration, growth).
- **KEY ASPECTS:**
 - a. **Optimum Temperature:** Every crop has an *ideal* temperature range.
 - **Tropical Crops (Warm-Season):** These are the base of Nigerian agriculture. They thrive in high temperatures (25-35°C).
 - *Examples: Yam, Cassava, Okra, Mango, Oil Palm.*
 - **Temperate Crops (Cool-Season):** These crops *require* cool conditions. High heat (e.g., 30°C+) causes them to "bolt" (flower prematurely) or die.
 - *Examples: Irish Potato, Cabbage, Carrots, Lettuce.*
 - **Ecological Niche:** This is why these temperate crops are grown commercially *only* on the **Jos Plateau** and **Mambilla Plateau**. The high altitude (elevation) creates a cool, temperate-like "micro-climate."

3. RAINFALL (WATER)

- **IMPORTANCE:** Water is the "lifeblood" of the plant.
 - It is a key reactant in **photosynthesis**.
 - It is the "transport system" (**transpiration**) that carries nutrients (dissolved in water) from the soil to the leaves.
 - It maintains **turgor pressure**, which is what keeps the plant standing upright and crisp (when a plant lacks water, it *wilts*).
- **KEY ASPECTS:**
 - a. **Amount:** Too little (drought) causes wilting and death.
 - b. **Distribution:** This is the key factor in Nigeria. Our "Wet Season" and "Dry Season" dictate the entire farming calendar. **Irrigation** (from 1st Term) is the technology used to overcome the dry season.
 - c. **Excess Water (Waterlogging):**

- This is just as bad as drought. It is a major problem in the heavy-rain South.
- If the soil is "waterlogged" (saturated), all the air pores in the soil are filled with water.
- The plant's roots *cannot breathe* (they need oxygen for respiration) and they will **suffocate and die**, leading to **Root Rot** (Week 3). This is why we use **raised beds** and **ridges**.

4. WIND

- **IMPORTANCE (NEGATIVE EFFECTS):**
 - a. **Physical Damage:** High winds can **break** stems, **shred** large leaves (like banana/plantain, reducing their photosynthetic ability), and cause "**lodging**" (pushing over tall crops like maize or rice).
 - b. **Water Loss:** Wind *increases* the rate of evaporation and transpiration, "drying out" the plant and soil much faster.
 - c. **Soil Erosion:** Wind can blow away dry, bare topsoil (a major problem in the North during the dry season).
- **IMPORTANCE (POSITIVE EFFECTS):**
 - a. **Pollination:** Wind is *essential* for the pollination of **wind-pollinated** crops (e.g., **Maize**, all grasses).
 - b. **Disease Reduction:** A gentle breeze provides good **air circulation**, which dries leaves faster and reduces the risk of **fungal diseases**.
- **MANAGEMENT:** Farmers plant **Windbreaks** (a row of trees or tall grasses) on the windward side of the farm to slow down strong winds.

5. HUMIDITY

- **DEFINITION: Relative Humidity (RH)** is the amount of moisture (water vapor) in the air, expressed as a percentage.
- **KEY ASPECTS (A "TRADE-OFF"):**

- **High Humidity (e.g., > 80%):**
 - *Location:* Common in Southern Nigeria (Lagos, Port Harcourt).
 - **Pro:** It *reduces* water loss (transpiration) from the plant's leaves.
 - **Con (CRITICAL):** It is the **#1 factor** that promotes the growth and spread of **FUNGAL DISEASES**. The leaves stay wet for a long time, allowing fungal spores to germinate.
- **Low Humidity:**
 - *Location:* Common in Northern Nigeria / Harmattan season.
 - **Pro:** Very *low* fungal disease pressure.
 - **Con:** Causes *high* water loss (transpiration). Plants will wilt very quickly if not irrigated.

EVALUATION

1. What is **photosynthesis**? What 3 environmental ingredients does it need?
2. What is "Photoperiodism"? What is the difference between a "Short-Day" and "Long-Day" plant?
3. Why are crops like Irish Potatoes grown in Jos, but not in Lagos? (Hint: Temperature).
4. What is "waterlogging"? Why is it dangerous for a plant's roots?
5. What is the *main positive effect* and the *main negative effect* of high humidity on crops?

ASSIGNMENT

1. Write a 250-word report on "**Rain-fed vs. Irrigated Agriculture.**" Why is irrigation a "game-changer" for farmers, especially in Northern Nigeria?
2. What is a **Windbreak**? Draw a simple diagram of a farm, showing the direction of the prevailing wind and the correct placement of a windbreak.
3. Define "Transpiration." How is it affected by (a) High Temperature, (b) High Humidity, and (c) High Wind?
4. What is "Lodging" in maize?
5. Define: Abiotic, Photoperiod, Vernalization, Turgor Pressure, Humidity, Transpiration.



WEEK 5: MIDTERM EXAMINATION

This week is dedicated to the Midterm Examination, which will cover all topics from Weeks 1–4.

- **Topics:**

- **Week 1:** Fertilizer (Organic vs. Inorganic), Manure (Types, Pros/Cons), NPK, Application Methods (Broadcasting, Ring, Foliar).
- **Week 2:** Weeds (Effects: competition), Weed Control (Cultural, Physical, Chemical, Herbicides). Pests (Biting/Chewing vs. Piercing/Sucking), Pest Control (Cultural, Biological, Chemical, Insecticides).
- **Week 3:** Crop Diseases, Pathogens (Fungi, Bacteria, Viruses, Nematodes), Symptoms (Wilt, Mosaic, Galls), Prevention (Resistant Varieties, Crop Rotation, Sanitation), Control (Fungicides, Roguing).
- **Week 4:** Environmental Factors: Light (Photosynthesis, Photoperiod), Temperature (Optimum, Temperate vs. Tropical), Rainfall (Waterlogging), Wind (Lodging, Pollination), Humidity (Disease).

- **Format:** The test will include definitions, short-answer questions, comparisons, and scenario-based questions (e.g., "A farmer's soil is too acidic. What should he do?", "A plant has yellow mosaic patterns. What is the likely cause, vector, and control?").

WEEK 6: FARM RECORDS IN HORTICULTURE

INTRODUCTION: WHY KEEP RECORDS?

Farming is a science. **Farming is also a business.** A **Farm Record** is a systematic and written document of all activities, observations, and financial transactions that happen on the farm.

A farmer who does not keep records is "farming blind." They are just *guessing*. They do not know *if* they are making a profit, *what* is causing problems, or *how* to plan for next year. A professional horticulturist *must* be a good record-keeper.



Production



Financial



Inventory



Farm Diary

IMPORTANCE OF FARM RECORDS

1. Business & Financial Management (Profit/Loss):

- This is the #1 reason. Records are the *only* way to know if you are making or losing money. You can calculate your **Profit** = (Total Income) - (Total Expenses).
- 2. **Securing Loans (Bank Access):**
 - A bank will **never** give a loan to a farmer who has no records. They need to see a **business plan**, a **financial history**, and **proof of production** to know you are a serious and safe investment.
- 3. **Planning & Budgeting:**
 - Records from *this* season (e.g., "I spent N20,000 on fertilizer and N10,000 on labor") are essential for creating a realistic budget for *next* season.
- 4. **Efficiency & Decision Making:**
 - Good records help you answer key questions:
 - "Which variety of tomato (UC-82b or Roma) gave me the highest yield?"
 - "Which fertilizer (Brand X or Y) worked best?"
 - "My harvest was low in Plot B. What went wrong?" (The farm diary shows: "Pest outbreak in Plot B").
- 5. **Farm Security:** A livestock inventory record (or equipment inventory) can be used to identify and prove ownership in case of theft.
- 6. **Inheritance/History:** Provides a clear history of the farm's performance and management for the next generation.

TYPES OF FARM RECORDS

There are four main categories of records that must be kept.

1. PRODUCTION RECORDS

- **Definition:** These are records of *what the farm produces* (the "yield"). It tracks the *output* of the farm.
- **Examples:**
 - "Crate of eggs collected per day"
 - "Number of kids (goats) born"

- "Litres of milk produced per cow"
- **"Baskets/kg of tomatoes harvested from Plot A"**
- "Number of seedlings sold from the nursery"
- **Purpose:** To track the *productivity* and *efficiency* of the farm. A sudden drop in this record (e.g., fewer eggs) is the first sign of a problem (e.g., disease).

2. FINANCIAL RECORDS

- **Definition:** These are all records of *money* (income and expenses).
- **Examples:**
 - **Expenditure Record:** A log of *all* money spent (e.g., "Oct 1: Bought 1 bag of NPK - N5,000," "Oct 3: Paid 2 workers - N4,000").
 - **Income (Sales) Record:** A log of *all* money received (e.g., "Oct 10: Sold 5 baskets of peppers - N10,000").
 - **Profit and Loss (P&L) Statement:** A summary (usually monthly or yearly) that calculates: **Total Income - Total Expenditure = Profit (or Loss)**.
 - **Receipts & Invoices:** Keep *all* physical copies of what you buy and sell.

3. INVENTORY RECORDS (ASSET REGISTER)

- **Definition:** A detailed list of all *assets* (items of value) that the farm owns.
- **This is divided into two types:**
 - a. **Non-Consumable Inventory:** Permanent tools and equipment.
 - *Examples:* "3 cutlasses," "1 knapsack sprayer," "1 wheelbarrow," "1 water pump." (Tracks quantity, date purchased, and condition).
 - b. **Consumable Inventory:** Items that are "used up" and need to be re-stocked.
 - *Examples:* "5 bags of fertilizer," "2kg of 'Roma' tomato seed," "100ml of pesticide."
- **Purpose:** To track assets, plan for re-stocking, and prevent theft.

4. FARM DIARY (OPERATIONS RECORD)

- **Definition:** A daily or weekly *journal* of all *activities* performed on the farm.
- **Examples:**
 - "March 15: Cleared the land for the tomato bed."

- "March 20: Transplanted seedlings."
- "April 1: Applied NPK fertilizer (ring method)."
- "April 5: Noticed leaf spots; sprayed with fungicide."
- "April 7: Heavy rain and wind, some plants lodged."
- **Purpose:** This is perhaps the *most valuable* record for a horticulturist. It helps you remember *what* you did and *when*, so you can **repeat successes** and **avoid failures** next year. It provides the *context* for the other records.

EVALUATION

1. Define "Farm Records" in your own words.
2. What is the *most important* reason for a farmer to keep financial records? (Hint: Banks).
3. List the 4 main types of farm records.
4. A farmer writes down "50kg of Urea, 10 cutlasses." What *type* of record is this?
5. A farmer writes down "May 10: Planted maize." What *type* of record is this?

ASSIGNMENT

1. Write a 250-word essay on the topic: "A Farmer Without Records is Farming Blind."
2. Choose *one* crop you would like to grow (e.g., tomatoes). Create a sample "Farm Diary" for the first 4 weeks of its life.
3. Create a simple "Financial Record" (Income and Expenditure) for a small poultry farmer for one week. Include at least 3 expenses (e.g., feed, water) and 2 sales (e.g., eggs, old layers).
4. What is a "Profit and Loss Statement"?
5. Define: Asset, Consumable, Inventory, Yield, P&L Statement.

WEEK 7: MIDTERM BREAK

This week is for the Midterm Break. There are no formal lesson notes.



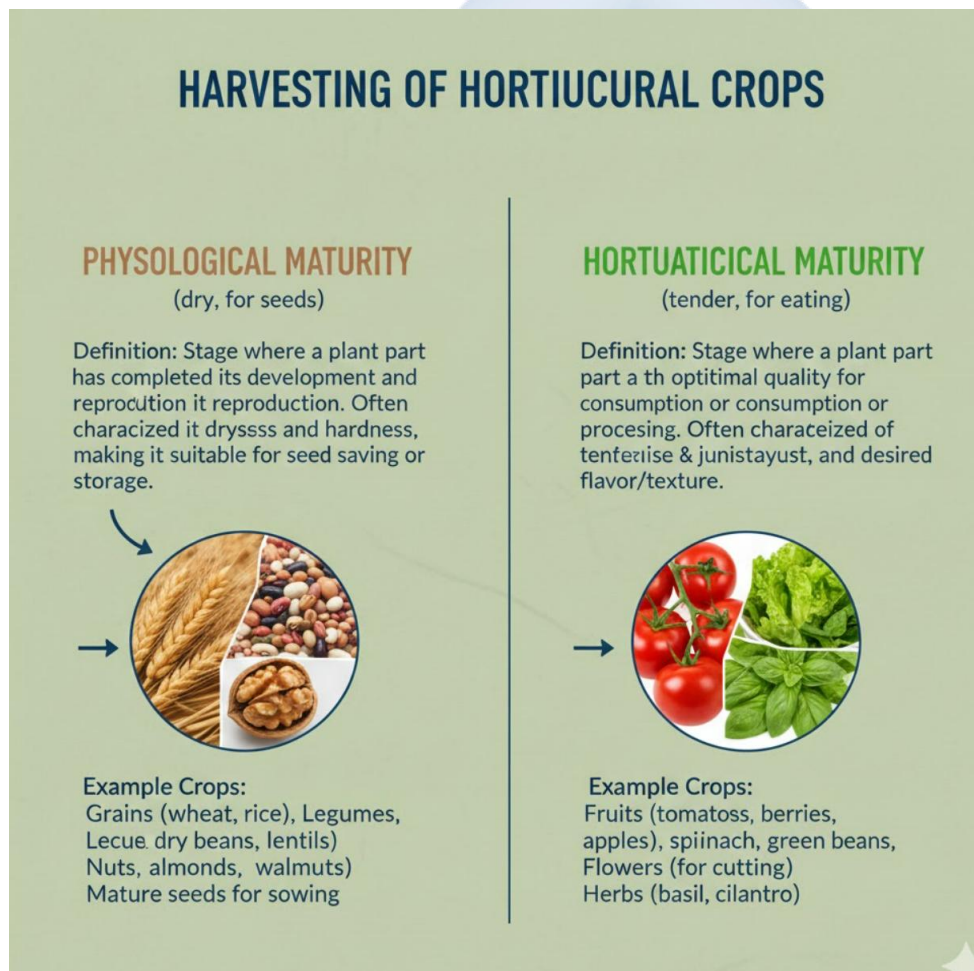
WEEK 8: HARVESTING OF HORTICULTURAL CROPS

🌿 INTRODUCTION TO HARVESTING

Harvesting is the process of gathering the mature crop from the field. It is the *reward* for all the previous work (planting, weeding, fertilizing).

This stage is *critical* because:

1. **Timing:** Harvesting at the wrong time (too early or too late) can ruin the quality of the crop.
2. **Method:** Harvesting incorrectly (roughly) can *damage* the crop, leading to rapid spoilage (rot) and economic loss.



STAGES OF MATURITY (WHEN TO HARVEST)

"Maturity" is the stage when the crop is ready for harvest. There are two types:

1. PHYSIOLOGICAL MATURITY:

- **Definition:** This is when the plant has *completed its full life cycle*. The seed is fully developed, and the plant may be "dying" (senescence).
- **Examples:**
 - **Maize:** Harvested when the cobs are dry, the leaves are brown, and the seeds are "hard."
 - **Rice:** Harvested when the grains are hard and dry.
 - Crops harvested for *dry seeds* (like beans) are harvested at this stage.

2. HORTICULTURAL (COMMERCIAL) MATURITY:

- **Definition:** This is the stage when the crop is at its *peak quality* for the *consumer*. This is often *before* physiological maturity. We harvest it when it is best for eating, not for its seed.
- **Examples:**
 - **Okra:** Must be harvested when young, tender, and "snappy." (If left to become physiologically mature, the pod becomes hard, woody, and inedible).
 - **Tomatoes:** Often harvested at the "**mature green**" or "pink" stage. They are *physiologically* immature (not fully red), but this allows them to be *transported* long distances (e.g., from Kano to Lagos) without becoming mushy. They will ripen *after* harvesting.
 - **Lettuce/Ugu:** Harvested when the leaves are young and tender, long before they flower.

MATURITY SIGNS (HOW TO TELL IT'S READY):

- **Color Change:** (e.g., Peppers turning from green to red, Tomatoes turning pink, Oranges turning orange).
- **Size & Shape:** (e.g., Cucumbers or Okra reaching a specific, usable size).

- **Ease of Separation:** (e.g., A ripe mango will detach from the stem easily).
- **Dryness:** (e.g., Onions are ready when their green tops fall over and turn brown).

HARVESTING TOOLS AND EQUIPMENT

The tool must match the crop to minimize damage.

- **HAND-PICKING (Using only hands):**
 - **Use:** For the most delicate crops (e.g., berries, green beans, leafy vegetables).
- **CUTTING TOOLS:**
 - **Knives:** A simple, sharp knife is used for harvesting leafy greens (like *Ugu* or *Ewedu*), cabbage, and lettuce.
 - **Secateurs (Pruners):** (From 1st Term, Wk 4). Used for a *clean cut* on fruits that have a tough stem (e.g., **Tomatoes (on the vine), Peppers, Eggplant, Oranges**). This prevents tearing the plant.
 - **Cutlass (Machete):** A heavy tool for harvesting large, tough crops (e.g., **Yams, Cassava, Plantain, Banana, Sugarcane**).
- **DIGGING TOOLS:**
 - **Garden Fork:** The *best* tool for lifting root crops (e.g., **Irish Potatoes, Carrots**) because the tines (prongs) loosen the soil *without* cutting the crop.
 - **Spade / Hoe:** Used for digging up yams and cassava.

METHODS OF HARVESTING & PRECAUTIONS

HARVESTING METHODS

1. **Hand-Picking / Snapping:** Pulling the fruit/pod from the plant (e.g., okra, beans).
2. **Cutting:** Using a knife or secateurs (e.g., tomatoes, peppers, cabbage).
3. **Lifting / Digging:** Uprooting root crops (e.g., carrots, yams).

HARVESTING PRECAUTIONS (CRITICAL FOR SHELF LIFE)

1. **HARVEST AT THE RIGHT TIME (OF DAY):**

- The *best* time to harvest all horticultural crops (especially fruits and leafy greens) is in the **cool, early morning**.
 - **Why?** The plant is full of water (**turgid**), and the "**field heat**" (retained heat from the sun) is low. A tomato harvested at 8 AM will last *much longer* than one harvested at 2 PM in the hot sun.
2. **HANDLE GENTLY (THE "EGG" RULE):**
- Treat all horticultural produce like *eggs*.
 - **A single bruise, cut, or fingernail puncture** is an "open wound" where **bacteria and fungi (rot)** will enter. This is the **#1 cause of post-harvest loss**.
 - Never "throw" produce into a basket. Place it gently.
3. **USE CLEAN TOOLS AND CONTAINERS:**
- Dirty knives or baskets covered in old mud can transfer disease (like bacteria) to the fresh produce.
4. **IMMEDIATE COOLING:**
- After harvesting, the produce must be moved to a **cool, shaded area** *immediately*. Do not leave the baskets sitting in the hot sun to "cook."
5. **HARVESTING DRY:** Do not harvest leafy greens (like *ugu*) when they are wet with rain or dew. The excess moisture will cause them to rot very quickly.

EVALUATION

1. What is the main difference between "Physiological Maturity" and "Horticultural Maturity"?
2. Give an example of a crop harvested at (a) Physiological Maturity and (b) Horticultural Maturity.
3. What is the *best* time of day to harvest leafy vegetables, and why?
4. What is the *best* hand tool for harvesting Irish potatoes, and why is it better than a spade?
5. List three essential "Harvesting Precautions" and explain *why* they are important.

ASSIGNMENT

1. Write a 250-word report on why "gentle handling" is the most important factor during harvesting. (You must use the words "bruise," "wound," "bacteria," "fungi," and "spoilage").
2. Go to a local market. Observe the tomatoes. Find an example of a "mature green" tomato and a "fully ripe" tomato. Why would a farmer harvest the green ones?
3. What is "field heat"?
4. Research the harvesting of Yams. Describe the manual process and the precautions taken.
5. Define: Harvesting, Maturity, Turgid, Field Heat, Senescence, Spoilage.



WEEK 9: POST-HARVEST HANDLING

🍎 INTRODUCTION: THE POST-HARVEST CHALLENGE

Post-Harvest Handling refers to *all* the operations that are performed on a crop *after* it has been harvested (e.g., sorting, grading, packaging, cooling, storage, and transport).

The Problem: This is the weakest link in Nigeria's food chain. It is estimated that **30-50%** of all horticultural crops (especially tomatoes, mangoes, and leafy greens) **spoil and are wasted** between the farm and the consumer.

The Goal: The goal of post-harvest handling is to **slow down spoilage** and **maintain the quality** of the crop, so it can be sold for the highest price.



1. SORTING

- **Definition:** This is the *first* and *most critical* step after harvest.
- **Process: Sorting** is the process of **separating the "bad" produce from the "good" produce**.
- **Action:** You must *remove* and *discard* any items that are:
 - Diseased (e.g., have rot, black spots)
 - Damaged (e.g., bruised, cut, punctured by insects)
 - Severely unripe or overripe.
- **The "One Bad Apple" Rule:** If you leave *one* rotten, diseased tomato in a basket, it will:
 - a. Release **ethylene gas** (a ripening hormone) that makes *all* other tomatoes ripen and rot faster.
 - b. Spread **fungal spores** and bacteria to its neighbors.
- Sorting *prevents* this chain reaction.

2. GRADING

- **Definition: Grading** is the next step. It is the process of **sorting the "good" produce into different categories** based on quality, size, and appearance.
- **Criteria:** Produce is graded by:
 - **Size:** (e.g., Large, Medium, Small)
 - **Color:** (e.g., Red tomatoes vs. Pink tomatoes)
 - **Shape:** (e.g., Straight cucumbers vs. Curved cucumbers)
 - **Freedom from Blemish:** (e.g., "Grade A" has perfect, flawless skin)
- **Importance:**
 - **Higher Price:** Graded produce **commands a higher price**. A uniform basket of "Grade A" tomatoes sells for *much more* than a mixed, ungraded basket.
 - **Market Access:** Supermarkets (like Shoprite), exporters, and food processors *will only buy* graded produce.

3. PACKAGING

- **Purpose:** Packaging is the container used to **protect** the produce from physical damage (bruising, crushing) during transport and storage.
- **TRADITIONAL (POOR) PACKAGING:**
 - **Examples:** Large, woven **rafia baskets** (common for tomatoes); large mesh bags; "over-stuffing" a car boot.
 - **Problems:**
 - i. **Bruising:** The rough, rigid basket walls damage the fruit.
 - ii. **No Ventilation:** Baskets are packed too tight, causing "field heat" to build up, which *accelerates* ripening and rot.
 - iii. **Compression:** The weight of the 20kg of tomatoes at the top *crushes* the 20kg at the bottom into a paste.
- **MODERN (IMPROVED) PACKAGING:**
 - **Examples:**
 - **Reusable Plastic Crates (RPCs):** These are the *best* solution. They are **rigid** (protect), **ventilated** (allow heat to escape), and **stackable** (the load is on the *crate*, not the fruit).
 - **Cardboard Boxes:** (Often with dividers, like for eggs or apples).
 - **Small Plastic Nets / Bags:** For small, pre-graded amounts for retail.

4. STORAGE

- **Purpose:** To preserve the crop and **extend its shelf life**, allowing the farmer to sell it later (when prices might be higher) or to transport it over long distances.
- **METHODS:**
 - a. **CURING:**
 - **Definition:** A "healing" process used for **root & tuber crops** (Yams, Cassava, Irish Potatoes).
 - **Process:** The crops are kept in a warm, humid, shaded place for 4-10 days.

- **Result:** This allows any small cuts or bruises on the skin to "heal" (this healing process is called **suberization**), forming a tough, new protective layer. This *dramatically* reduces rot in long-term storage.

b. **COLD STORAGE (REFRIGERATION):**

- **Definition:** The *best* method for most fruits and vegetables (e.g., tomatoes, peppers, leafy greens).
- **Process:** Storing produce in a **cold room** or refrigerator (at 4-10°C).
- **Result:** The cold temperature **slows down** all life processes: it slows ripening, slows water loss (wilting), and slows the growth of bacteria and fungi.

c. **VENTILATED STORAGE (TRADITIONAL):**

- **Definition:** Storing crops that do *not* like cold, but *do* need good airflow.
- **Examples:**
 - **Yam Barn:** A traditional structure of poles (a *rhombus*) where yams are tied individually to allow 360° airflow, preventing rot.
 - **Onion/Garlic:** Stored in *net bags* or on *racks* in a dry, ventilated shed. (If you refrigerate an onion, it will sprout and become soft).

EVALUATION

1. What is the main difference between **Sorting** and **Grading**?
2. What is the "One Bad Apple" rule, and why is sorting so important? (Hint: Ethylene gas).
3. List three major *problems* with using traditional *rafia* baskets for transporting tomatoes.
4. What are the three main advantages of using **Reusable Plastic Crates (RPCs)**?
5. What is "**Curing**," and which crops is it used for? What is the "healing" process called?

ASSIGNMENT

1. Write a 250-word report on the "post-harvest losses" in the Nigerian tomato industry. (How much is lost, and *why*?).
2. Draw a diagram of a traditional **Yam Barn (Rhombus)**.

3. What is a "cold chain"? (Research this term). Why is an *unbroken* cold chain essential for exporting fruits?
4. You are a farmer who has just harvested 100 baskets of "Grade A" tomatoes and 50 baskets of "Grade C" tomatoes (small, misshapen). How would your *marketing plan* (who you sell to) be different for these two grades?
5. Define: Post-Harvest, Shelf Life, Curing, Suberization, Cold Chain, RPC.



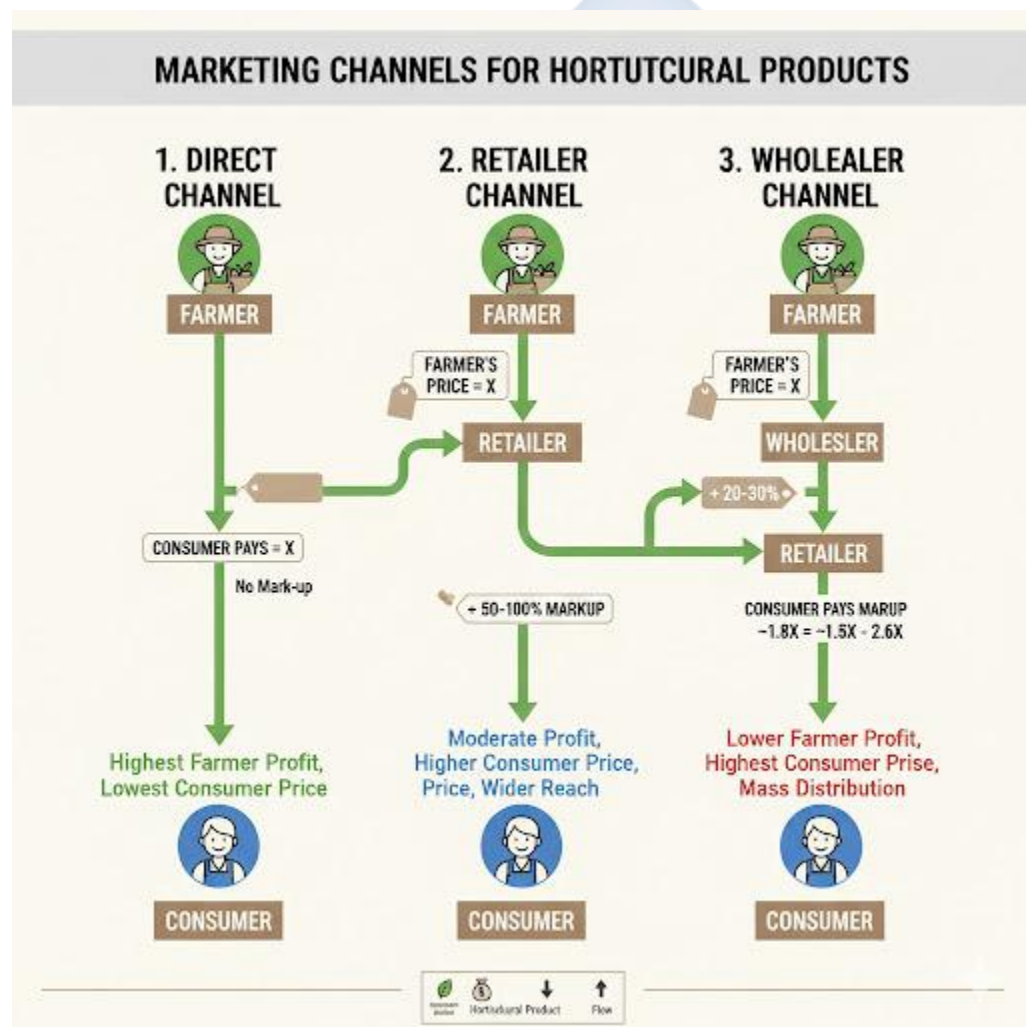
WEEK 10: MARKETING OF HORTICULTURAL PRODUCTS

INTRODUCTION TO MARKETING

Marketing is *not* just "selling." It is the entire chain of activities and decisions that connects the **farm** (the producer) to the **consumer's table**.

For horticultural crops, marketing is especially challenging due to two factors we've learned:

1. **High Perishability:** They rot quickly.
2. **Seasonality:** They all get harvested at the same time.



MARKETING CHANNELS

This is the "path" the product takes from the farmer to the final consumer.

1. DIRECT MARKETING (Farmer → Consumer):

- **Method:** The farmer sells *directly* to the public. There is no "middleman."
- **Examples:** Selling at a **roadside stand**, a "U-Pick" farm (where customers pick their own), or a **Farmer's Market**.
- **Pros:** The farmer gets **100% of the profit** and builds a relationship with customers.
- **Cons:** Very time-consuming; only sells a small volume.

2. RETAIL MARKETING (Farmer → Retailer → Consumer):

- **Method:** The farmer sells their produce to a retailer (like a market woman, a hotel, a restaurant, or a supermarket).
- **Examples:** A farmer has a contract to supply 10 baskets of peppers to a local Mr. Bigg's every week, or to supply 20 crates of lettuce to Shoprite.
- **Pros:** Good, stable, and predictable income.
- **Cons:** Retailers (especially supermarkets) demand *very high quality* and *graded* produce.

3. WHOLESALE MARKETING (Farmer → Wholesaler → Retailer → Consumer):

- **Method:** This is the *most common* channel for bulk producers. The farmer sells their *entire* harvest (e.g., a truckload) to a **Wholesaler (Middleman)** at the farm gate or a large collection market (like Mile 12 in Lagos or Onitsha Main Market).
- **Pros:** **Fast, easy, and low-risk** for the farmer. They sell 1,000 baskets in *one* transaction and get paid immediately.
- **Cons:** The farmer gets the **lowest possible price** per unit. The middleman (wholesaler) takes on the risk of transport and spoilage, but also takes the largest share of the profit.

PRICING OF HORTICULTURAL PRODUCTS

What determines the price of a basket of tomatoes? It's a classic case of **Supply and Demand**.

- **Cost of Production:** The *base* price. The farmer must cover the cost of seeds, fertilizer, labor, etc.
- **Seasonality (Supply and Demand):** This is the **most powerful factor** in Nigeria.
 - **Harvest Season (The "Glut"):** *All* farmers are harvesting at the same time (e.g., April for mangoes, March for tomatoes). There is a *glut* (over-supply) of produce. Supply is high, demand is stable, so **prices crash**. (This is bad for the farmer).
 - **Off-Season (The "Scarcity"):** It is the dry season, and only farmers with *irrigation* can grow. Supply is low, demand is high, so **prices are very high**. (This is bad for the consumer).
- **Quality (Grading):** A graded, clean, uniform basket of produce will always sell for more than a mixed, dirty, or bruised basket.
- **Transportation Cost:** The cost of fuel and transport to get the produce from the farm (e.g., Kano) to the city market (e.g., Lagos) is added to the price.

MAJOR CHALLENGES OF HORTICULTURAL MARKETING IN NIGERIA

1. PERISHABILITY & POOR STORAGE (The #1 Problem):

- **Problem:** Because crops like tomatoes and mangoes rot in 2-3 days, the farmer has **no bargaining power**. They *must* sell to the first middleman who shows up, at *any* price, or risk losing 100% of their harvest.
- **Solution: Cold Storage (Cold Chain)** facilities.

2. POOR TRANSPORTATION:

- **Problem:** Bad roads, slow trucks, and hot, unventilated vehicles mean that a large portion of the produce is damaged *in-transit* (bruised, vibrated, rotted).
- **Solution:** Improved roads, and **Reusable Plastic Crates (RPCs)** instead of *rafia* baskets.

3. SEASONALITY (GLUT & SCARCITY):

- **Problem:** The "boom and bust" cycle where prices crash during harvest (bad for farmers) and skyrocket in the off-season (bad for consumers).
- **Solution: Irrigation** (to grow in the off-season) and **Processing (Value Addition)** (e.g., turning the cheap tomatoes during the glut into high-value, shelf-stable *tomato paste*).

4. INEFFICIENT SUPPLY CHAIN (MIDDLEMEN):

- **Problem:** The supply chain is long and inefficient, with many "middlemen" each taking a cut. This means the *farmer* gets a very low price, and the *consumer* pays a very high price.
- **Solution:** Farmer Cooperatives, better logistics.

EVALUATION

1. What is "marketing" in the context of horticulture?
2. Describe the three main "marketing channels." Which one gives the farmer the *highest* profit per item?
3. What is a "Wholesaler (Middleman)"? What is the main "pro" and "con" of using one for the farmer?
4. What is "**Seasonality**"? What happens to the price of mangoes during the harvest season (the "glut")?
5. List three of the *major challenges* of horticultural marketing in Nigeria.

ASSIGNMENT

1. Write a 250-word essay on the problem of **Perishability**. How would a "cold storage" (cold chain) facility revolutionize the tomato industry in Nigeria?
2. Draw a simple flowchart illustrating the three main marketing channels (Direct, Retail, Wholesale).
3. What is "Value Addition"? How does turning a fresh tomato into "dried tomato" or "tomato paste" solve the problem of perishability?

4. Visit a local market. Ask a tomato seller how much a basket of tomatoes costs *today*, and how much it cost *3 months ago*. Write a short report on your findings and explain the price difference.
5. Define: Glut, Scarcity, Middleman, Farm-gate Price, Perishability, Supply Chain.



WEEK 11: REVISION

GENERAL REVIEW OF ALL SECOND TERM TOPICS

This week is for a comprehensive review of all topics covered this term. We have followed the entire journey of a crop, from feeding it in the ground to selling it at the market.

- **WEEK 1: FERTILIZER & MANURE**

- **Organic:** (Manure, Compost). Improves *soil structure*, slow-release, bulky.
- **Inorganic:** (NPK, Urea). *Fast-acting*, concentrated, "burns" if over-used, *does not* improve soil structure.
- **NPK:** Nitrogen (leaves), Phosphorus (roots/fruits), Potassium (vigor).
- **Methods:** Broadcasting (all over), Ring (trees), Side Placement (rows), Foliar (leaf spray).

- **WEEK 2: WEEDS & PESTS**

- **Weeds:** Compete for **Light, Water, Nutrients, Space**.
- **Weed Control:** Cultural (Mulching, Spacing), Physical (Hoeing), Chemical (**Herbicides** - Selective vs. Non-selective).
- **Pests:** Biting/Chewing (holes in leaves) vs. Piercing/Sucking (suck sap, spread **viruses**).
- **Pest Control:** Cultural (Crop Rotation), Biological (Ladybugs), Chemical (**Pesticides**).

- **WEEK 3: CROP DISEASES**

- **Pathogens:** **Fungi** (most common, love humidity), **Bacteria** (need wounds), **Viruses** (spread by insect vectors, *no cure*), **Nematodes** (soil worms, attack roots).
- **Symptoms:** Wilt, Mosaic, Galls, Damping-Off.
- **Prevention:** **Resistant Varieties, Crop Rotation, Sanitation** (Roguing).
- **Control:** Fungicides, Nematicides, Insecticides (to kill vectors).

- **WEEK 4: ENVIRONMENTAL FACTORS**

- **Light:** For **Photosynthesis**. *Duration* (Photoperiod) triggers flowering.

- **Temperature:** Controls growth rate (e.g., cool-season crops like potatoes in Jos).
- **Rainfall:** Too much = **Waterlogging** (roots suffocate). Too little = wilt.
- **Wind:** Causes **Lodging** (damage) but is needed for **Pollination** (maize).
- **Humidity:** High humidity = high **Fungal Disease** risk.
- **WEEK 6: FARM RECORDS**
 - **Importance:** To get **bank loans**, calculate profit, and make decisions.
 - **Types:** **Production** (yields), **Financial** (income/expense), **Inventory** (tools/supplies), **Farm Diary** (activities).
- **WEEK 8: HARVESTING**
 - **Maturity:** Physiological (dry seed) vs. **Horticultural** (e.g., tender okra, "mature green" tomato).
 - **Time:** Harvest in the **cool, early morning** to reduce "field heat."
 - **Precautions:** **Handle gently** (like eggs) to prevent bruising, which leads to rot.
- **WEEK 9: POST-HARVEST HANDLING**
 - **Sorting:** Removing the *bad* produce. (Prevents "One Bad Apple").
 - **Grading:** Grouping the *good* produce by size/color (increases price).
 - **Packaging:** **Plastic Crates** (good) vs. *Rafia Baskets* (bad).
 - **Storage:** **Curing** (for yams/potatoes), **Cold Storage** (for tomatoes/greens), **Barns** (for yams).
- **WEEK 10: MARKETING**
 - **Channels:** Direct (farmer's market), Retail (supermarket), Wholesale (middlemen).
 - **Challenges:** **Perishability** (no bargaining power), **Seasonality** (glut/scarcity price crashes), **Transportation** (bad roads).

EVALUATION

1. What is the main difference between organic and inorganic fertilizers in terms of their effect on *soil structure*?
2. A farmer sees his tomato leaves are covered in a white, powder-like substance. What *type* of pathogen (Fungus, Virus, or Bacteria) is the most likely cause?
3. What is "Photoperiodism"?
4. Why is "Hardening Off" a critical step before transplanting?
5. What is the *single biggest challenge* in marketing horticultural crops in Nigeria, and what is its main cause?

ASSIGNMENT

1. **Project:** You are a small farmer in Nigeria. You want to grow one hectare of high-quality **tomatoes** for sale. Write a one-page "Production Plan" that addresses:
 - (a) How you will manage the soil's fertility (Week 1).
 - (b) The main disease you will face and your prevention strategy (Week 3).
 - (c) The irrigation method you will choose and why (from 1st Term).
 - (d) The post-harvest packaging you will use and why (Week 9).
 - (e) The marketing channel you will choose (Week 10).
2. Draw a simple flowchart illustrating the three main marketing channels (Direct, Retail, Wholesale).
3. What is "Value Addition"? How does turning a fresh tomato into "dried tomato" or "tomato paste" solve the problem of perishability?
4. Create a sample "Farm Diary" for one week of managing a vegetable nursery.
5. Define the 10 most important terms you learned this term.

SS1 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

THIRD TERM

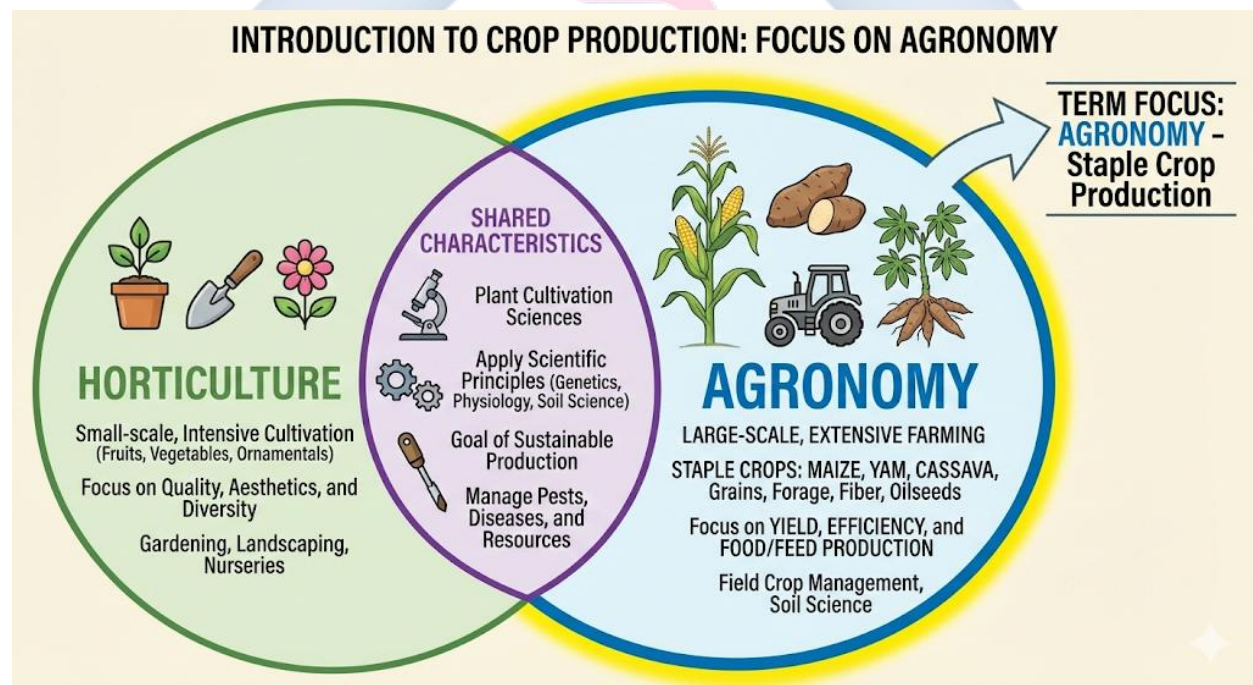
WEEK 1: INTRODUCTION TO CROP PRODUCTION

🌿 DEFINITION OF CROP PRODUCTION

Crop production is a major branch of agriculture that involves the art, science, and business of **cultivating plants (crops)** for human and animal use. It is a deliberate and systematic process that includes land preparation, planting, managing, harvesting, and storing plants to produce:

- **Food** (e.g., grains, tubers, fruits, vegetables)
- **Fiber** (e.g., cotton)
- **Fuel** (e.g., sugarcane for ethanol, jatropha for biodiesel)
- **Medicine** (e.g., lemongrass, neem)
- **Animal Feed** (e.g., fodder, maize)

It is one of the most fundamental human activities, forming the basis of all settled civilizations and economies.



DISTINCTION: HORTICULTURE VS. AGRONOMY

"Crop Production" is a broad, parent term that covers two main specialized fields: Horticulture (which we introduced in the First Term) and Agronomy.

Feature	HORTICULTURE (Term 1)	AGRONOMY (Term 3 Focus)
Definition	"Garden Cultivation"	"Field Cultivation"
Scale	Intensive (high input, high value per plant/area)	Extensive (large scale, e.g., hectares)
Crops	Fruits, vegetables, flowers, herbs, ornamental plants	Grains, cereals, tubers, fibers (staple/field crops)
Examples	Tomatoes, Mangoes, Okra, Roses, Carpet Grass	Maize (Corn), Rice, Cassava, Yams, Soybeans, Cotton
Goal	High quality, aesthetics, and high value (often perishable)	High yield (bulk) and long-term storage (often dry)

This term, our focus on "Crop Production" will lean heavily on **agronomic crops**, which are the staples of our food supply.

IMPORTANCE OF CROP PRODUCTION IN NIGERIA

Crop production is the **backbone** of the Nigerian economy and the **foundation** of its national food security.

1. FOOD SECURITY (PRIMARY IMPORTANCE):

- **Provision of Staples:** Crop production provides the staple foods that form the core of the Nigerian diet. These are the main sources of **calories (energy)** for over 200 million people.
 - **Roots/Tubers:** **Cassava** (for *Eba/Garri, Fufu, Lafun*), **Yams** (pounded yam, boiled), Potatoes.
 - **Cereals:** **Rice** (*Jollof, Tuwo*), **Maize** (Corn - for *Ogi, Pap, Tuwo*), **Millet, Sorghum**.
- **Nutritional Value:** It provides all the "soup and stew" ingredients (tomatoes, peppers, onions, *ugu*) which are the main sources of **vitamins and minerals** (as learned in horticulture).

2. ECONOMIC IMPORTANCE:

- **Employment:** This is the **largest employer** in the Nigerian economy. An estimated 60-70% of the entire labor force is involved in some form of agriculture, primarily subsistence and small-scale crop production.
- **Gross Domestic Product (GDP):** Agriculture (with crop production as its largest component) is a massive contributor to Nigeria's GDP, representing the most significant part of the non-oil economy.
- **Foreign Exchange Earnings (Exports):** It is a major source of income from other countries.
 - **Traditional Cash Crops:** Nigeria is a world-leading exporter of **Cocoa** (for chocolate), **Oil Palm** (for palm oil), **Cashew nuts**, and **Rubber**.
 - **Horticultural Cash Crops:** Nigeria also exports **Hibiscus (zobo)**, **Ginger**, **Sesame Seeds**, and garlic.
- **Raw Materials for Industries:** Crop production supplies the raw materials that fuel other "agro-allied" industries (the "value chain").
 - **Textile Industry:** Cotton for fabric.
 - **Food Processing:** Tomatoes for paste, oranges for juice, cassava for starch, maize for cornflakes and animal feed.
 - **Pharmaceutical:** Medicinal crops for drugs.

3. SOCIAL AND CULTURAL IMPORTANCE:

- **Poverty Alleviation:** It provides a direct livelihood and source of food for the majority of the rural population.
- **Cultural Heritage:** Many crops are central to our culture and traditions (e.g., the **New Yam Festival** in Igbo land, the use of **Kola nut** in ceremonies and as a sign of respect).

EVALUATION

1. Define "Crop Production" in your own words.

2. What is the main difference in *scale* and *crop type* between **Horticulture** and **Agronomy**?
3. List the three main *economic* importances of crop production in Nigeria.
4. What is a "staple food"? Name three major staple crops in Nigeria.
5. What is a "cash crop"? Name two traditional and two horticultural cash crops exported by Nigeria.

ASSIGNMENT

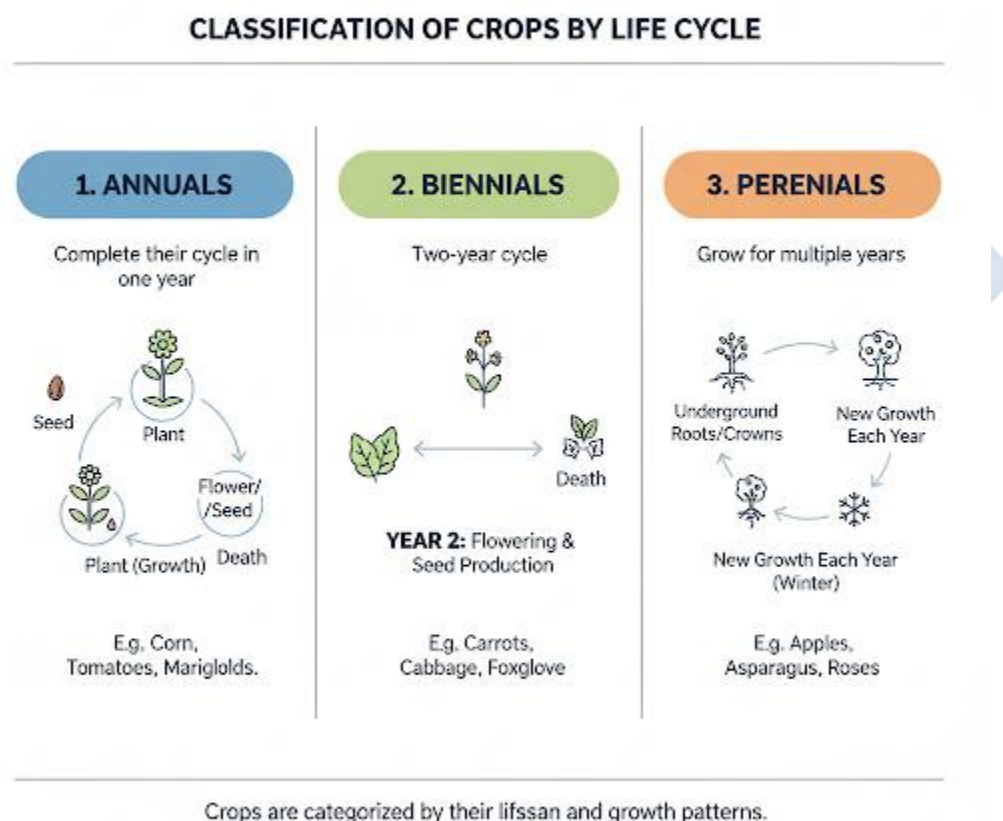
1. Write a 250-word essay on the topic: "Without crop production, Nigeria's economy and food security would collapse."
2. List five (5) industries in Nigeria and the *specific crop* they depend on as a raw material (e.g., "Tire Industry - Rubber").
3. What is "Food Security"? Explain in one paragraph how crop production is essential for achieving it.
4. Define the following terms: Agronomy, GDP, Staple Crop, Foreign Exchange, Value Chain, Agro-allied.
5. Interview a family member. Ask them to list all the food they ate yesterday. Create a table showing which of those foods were a *direct product* of crop production (e.g., Eba ← Cassava; Bread ← Wheat).

WEEK 2: CLASSIFICATION OF CROPS BY LIFE CYCLE

✿ INTRODUCTION TO LIFE CYCLES

One of the most important ways to classify plants is by their **life cycle**. The life cycle is the time a plant takes to go from a **seed (germination)**, to a mature plant (vegetative growth), to a **flower (reproduction)**, to a **seed (production)**, and finally to its **death**.

This classification is critical for the farmer, as it determines *when* a crop will be profitable, *how* it must be managed, and *how often* it must be planted.



1. ANNUAL CROPS

- **Definition:** An **annual** is a plant that completes its *entire* life cycle, from seed germination to producing its own new seeds, in **one single growing season** (which is less than one year). After producing its seeds, the parent plant dies.
- **Characteristics:**
 - They grow rapidly (fast life cycle).
 - They are often "heavy feeders" (require fertile soil to support their fast growth).
 - The farmer must **replant them from seed *every single season***.
- **Categories & Examples:**
 - **Cereals:** Maize (Corn), Rice, Millet, Sorghum, Wheat.
 - **Legumes (Pulses):** Beans, Soybeans, Groundnuts (Peanuts), Cowpea.
 - **Most Vegetables (Olericulture):** Tomatoes, Peppers, Okra, *Ugu* (Fluted Pumpkin), *Ewedu*, Spinach, Lettuce, Watermelon, Cucumber.

2. BIENNIAL CROPS

- **Definition:** A **biennial** is a plant that requires **two growing seasons (two years)** to complete its life cycle. The name comes from *bi-* (meaning "two") and *annus* (meaning "year").
- **Characteristics (The Two-Year Cycle):**
 - **YEAR 1 (Vegetative Growth):** The plant germinates from seed and spends the *entire first year* focusing on "vegetative" growth (growing strong leaves and a large root system). It stores a large amount of food (sugar/starch) in its root or leaves to survive the dormant season (winter or a long dry season).
 - **YEAR 2 (Reproductive Growth):** Using the stored energy from Year 1, the plant "bolts" (sends up a tall flower stalk), produces flowers, sets seeds, and then *dies*.
- **THE HORTICULTURAL CONFUSION:**

- For farmers, most biennials are grown as **annuals**. We *harvest* them in **Year 1** to eat the part with the *stored food*. We do not let them go to Year 2 because the "bolting" process turns the storage part bitter, woody, and inedible.
- **Examples:**
 - **Carrot:** We harvest the sweet taproot in Year 1. If we left it, it would flower in Year 2.
 - **Onion:** We harvest the bulb (storage leaves) in Year 1.
 - **Cabbage:** We harvest the head (leaves) in Year 1.
 - **Parsley:** We harvest the leaves in Year 1 (in Year 2, it flowers and the leaves are bitter).

3. PERENNIAL CROPS

- **Definition:** A **perennial** is a plant that lives for **more than two years**. It grows, flowers, and produces seeds *year after year* without dying at the end of the season.
- **Characteristics:**
 - They do not need to be replanted each year (they are "permanent").
 - They often have a long "**juvenile**" **period** (they take several years *before* they can produce their first fruit).
 - They are often **woody** (e.g., trees and shrubs).
 - They require long-term investment and management (e.g., pruning).
- **Categories & Examples:**
 - **All Tree Crops:** **Mango, Orange (all Citrus), Cocoa, Oil Palm, Cashew, Rubber, Kola nut, Avocado.**
 - **Herbaceous Perennials** (The "tops" die back, but the root lives): **Banana, Plantain, Sugarcane, Pineapple, Ginger.**
 - **Forage Crops:** (Permanent pasture grasses) **Elephant Grass, Guinea Grass.**
 - **Note:** Some crops, like **Cassava**, can be grown as an annual (harvested in 8-12 months) or a perennial (left for 2+ years, though the tuber becomes woody).

COMPARISON TABLE

Feature	ANNUAL	BIENNIAL	PERENNIAL
Life Span	One season (< 1 year)	Two seasons (2 years)	Many years (> 2 years)
Reproduction	Flowers & dies once	Flowers & dies in Year 2	Flowers <i>every year</i>
Planting	Must be replanted every season	Replanted every 1 or 2 years	Planted once for many years
Examples	Maize, Rice, Tomato, Okra	Cabbage, Carrot, Onion	Mango, Cocoa, Orange, Banana

EVALUATION

1. Define "Annual" and give three examples of annual crops grown in Nigeria.
2. Define "Biennial." Explain the plant's life in **Year 1** versus **Year 2**.
3. Why do farmers *harvest* most biennial crops (like carrots and onions) in their *first* year?
4. What is a "Perennial"? Give three examples of perennial *tree crops*.
5. Classify the following crops by their life cycle: (a) Rice, (b) Cabbage, (c) Mango, (d) Okra, (e) Onion.

ASSIGNMENT

1. Write a 200-word essay explaining the *management differences* a farmer would face when farming an annual crop (like maize) versus a perennial crop (like cocoa).
2. Go outside and find three different plants in your school compound or home. For each plant, (a) name it (or describe it), (b) take a leaf sample or draw it, and (c) classify it as an annual, biennial, or perennial, explaining your reasoning.
3. What does it mean for a plant to "bolt"? Which life cycle is this associated with?
4. Define the terms: Life Cycle, Germination, Vegetative Growth, Reproductive Growth, Juvenile Period, Bolting.
5. Create a "Farming Calendar" for an annual crop (like maize) showing the 12 months and when you would (a) Plant, (b) Weed/Fertilize, and (c) Harvest.



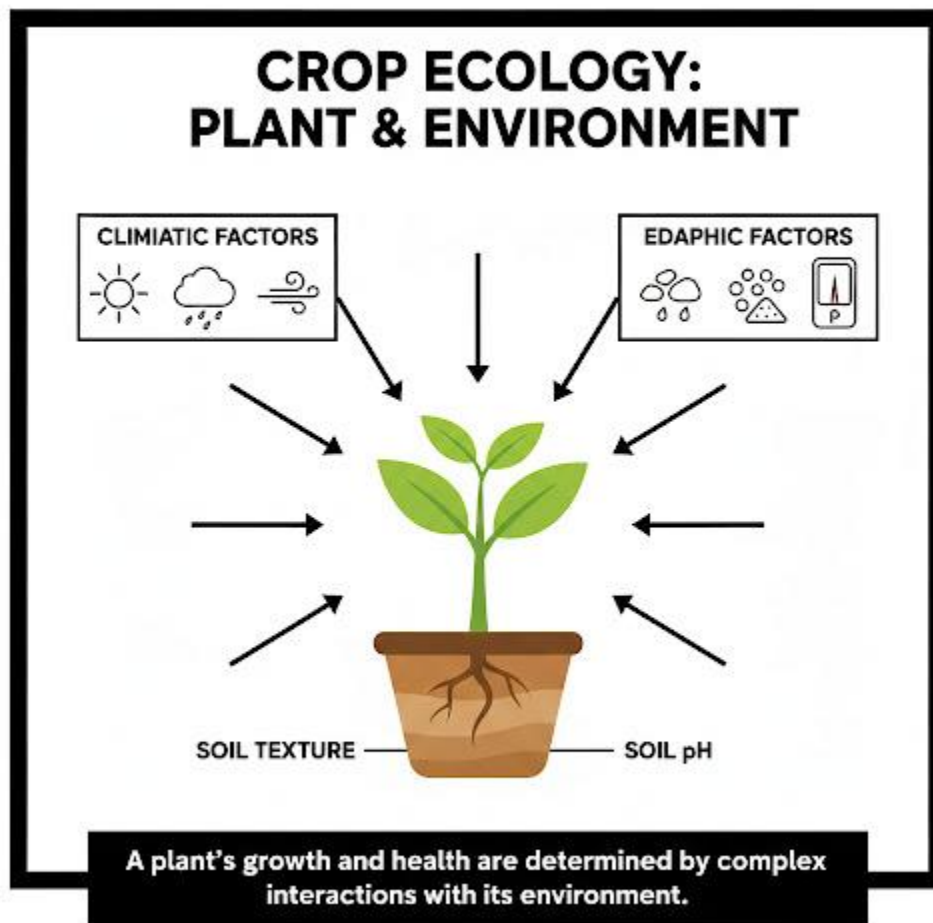
WEEK 3: CROP ECOLOGY

🌐 INTRODUCTION TO CROP ECOLOGY

Ecology is the study of how living organisms (like plants) interact with each other and with their *non-living environment*.

Crop Ecology, therefore, is the science of understanding the **environmental requirements** of a crop. It seeks to answer the question: "What is the *perfect* environment (climate and soil) for this specific crop to grow and produce the highest yield?"

Understanding this is the key to successful farming. You cannot "force" a crop to grow where it does not "belong." (You cannot grow a mango in the desert, and you cannot grow a date palm in the swamp).



The environmental factors are divided into two groups:

1. **Climatic Factors** (the air/weather)
2. **Edaphic Factors** (the soil)

CLIMATIC FACTORS (THE "ATMOSPHERE")

This is the most important set of factors, as a farmer has almost no control over them.

1. RAINFALL (WATER)

- **Importance:** Water is the "lifeblood" of the plant.
 - It is a key ingredient in **photosynthesis**.
 - It is the "transport system" (**transpiration**) that carries nutrients (dissolved in water) from the soil to the leaves.
 - It maintains **turgor pressure**, which is what keeps the plant standing upright and crisp (when a plant lacks water, it *wilts*).
- **Requirements:**
 - **Amount:** Crops vary wildly. **Rice** (paddy) needs *flooded* conditions. **Maize** needs moderate, consistent rain. **Millet** and **Sorghum** are *drought-resistant* and can survive on very *little* rain (making them ideal for Northern Nigeria).
 - **Distribution:** This is crucial. **Yams** need heavy rain at the *beginning* of the season (for tuber development) but *dryness* at the *end* (for harvesting).

2. TEMPERATURE (HEAT)

- **Importance:** Temperature controls the *rate* of all chemical reactions (like photosynthesis) and growth.
- **Requirements:**
 - **Tropical Crops (Warm-Season):** Most Nigerian staples are tropical. They *require* high temperatures (25-35°C) to thrive.
 - *Examples:* **Yam, Cassava, Okra, Mango, Oil Palm.**
 - **Temperate Crops (Cool-Season):** These crops *require* cool conditions. High heat (e.g., 30°C+) causes them to "bolt" (flower prematurely) or die.

- *Examples:* **Irish Potato, Cabbage, Carrots, Lettuce.**
- **Ecological Niche:** This is why these temperate crops are grown commercially *only* on the **Jos Plateau** and **Mambilla Plateau**. The high altitude (elevation) creates a cool, temperate-like "micro-climate."

3. LIGHT (SOLAR RADIATION)

- **Importance:** This is the *energy source* for **photosynthesis**.
- **Requirements:**
 - **Light Intensity (Brightness):**
 - **Sun-loving:** Most field crops (Maize, Rice, Sugarcane) need *full, bright sun* all day for high yield.
 - **Shade-tolerant:** Some horticultural crops *prefer or require* shade, especially when young. (e.g., **Cocoa, Coffee**).
 - **Light Duration (Photoperiodism):**
 - **Definition:** This is the biological *trigger* for **flowering** in many plants, based on the length of the day.
 - **Short-Day Plants:** Only flower when the days are *short* (less than 12 hours of light). (e.g., **Hibiscus / Zobo**, Soybean, many traditional rice varieties).
 - **Long-Day Plants:** Only flower when days are *long* (e.g., Cabbage, Lettuce, Carrots—this is why they don't flower and produce seeds easily in Nigeria).
 - **Day-Neutral Plants:** Flower based on *age* (maturity), not day length. (e.g., **Tomato**, most modern hybrid maize).

4. WIND

- **Positive Effects:**
 - a. **Pollination:** Wind is *essential* for the pollination of **wind-pollinated** crops (e.g., **Maize**, all grasses).
 - b. **Disease Reduction:** A gentle breeze provides good **air circulation**, which dries leaves faster and reduces fungal disease.

- **Negative Effects:**
 - a. **Physical Damage:** High winds can **break** stems, **shred** large leaves (like banana/plantain), and cause "**lodging**" (pushing over tall crops like maize or rice).
 - b. **Water Loss:** Wind *increases* the rate of evaporation and transpiration, "drying out" the plant and soil much faster.
 - c. **Soil Erosion:** Wind can blow away dry, bare topsoil (a major problem in the North during the dry season).

EDAPHIC FACTORS (THE "SOIL")

This refers to the soil requirements of the crop.

1. SOIL TEXTURE

- **Definition:** The percentage of **Sand, Silt, and Clay** (from 1st Term).
- **Requirements:**
 - **Sandy-Loam:** (Well-drained, loose). This is ideal for **Root & Tuber Crops** (Yam, Cassava, Carrots). They need *loose* soil so the tubers can expand easily.
 - **Clay/Clay-Loam:** (Holds water). This is ideal for crops that need a lot of water, like **Rice**.
 - **Loam:** (The "best" of both). Ideal for most vegetables (Tomatoes, Peppers).

2. SOIL PH (ACIDITY)

- **Importance:** Most crops prefer a **slightly acidic to neutral soil (pH 6.0 - 7.0)**.
- **Problem:** Most soils in the *humid South* of Nigeria are highly *acidic* (pH 4.5-5.5) due to heavy rainfall (which leaches out "basic" minerals like calcium). This acidity *locks up* nutrients like Phosphorus, making them unavailable to the plant.
- **Solution:** Farmers in the South must apply **Lime** to "sweeten" (raise the pH) of the soil.

3. SOIL FERTILITY

- **Importance:** Crops need NPK and other nutrients.
- **Requirements:**

- **"Heavy Feeders"**: Crops that remove a *lot* of nutrients from the soil. (e.g., **Maize, Yams, Tomatoes**). They *must* be heavily fertilized.
- **"Light Feeders"**: (e.g., Carrots, Onions).
- **"Soil Builders"**: **Legumes** (e.g., Beans, Cowpea, Groundnut). They have "nodules" on their roots that contain bacteria. These bacteria "fix" nitrogen from the air, making the soil *more* fertile.

EVALUATION

1. What is "Crop Ecology"?
2. What are the two main groups of environmental factors (Climatic and Edaphic)?
3. Explain why "Temperate" crops like Irish Potatoes are grown on the Jos Plateau.
4. What is "Photoperiodism"? Give one example of a "Short-Day" plant.
5. What is the *ideal* soil texture for (a) Rice and (b) Yams?

ASSIGNMENT

1. Write a 250-word essay on the *climatic* and *edaphic* factors that make the *North* of Nigeria ideal for growing Millet and the *South* ideal for growing Oil Palm.
2. What is a "Legume"? Draw a simple diagram of a bean root, showing the **Nitrogen-fixing nodules**.
3. Define the terms: Edaphic, Abiotic, Photoperiod, Lodging, Leaching (of soil).
4. Explain the difference between a "sun-loving" plant and a "shade-tolerant" plant, with one example of each.
5. A farmer in Asaba (humid South) wants to plant Cabbage. What two major *ecological challenges* (one climatic, one soil-related) will he face?

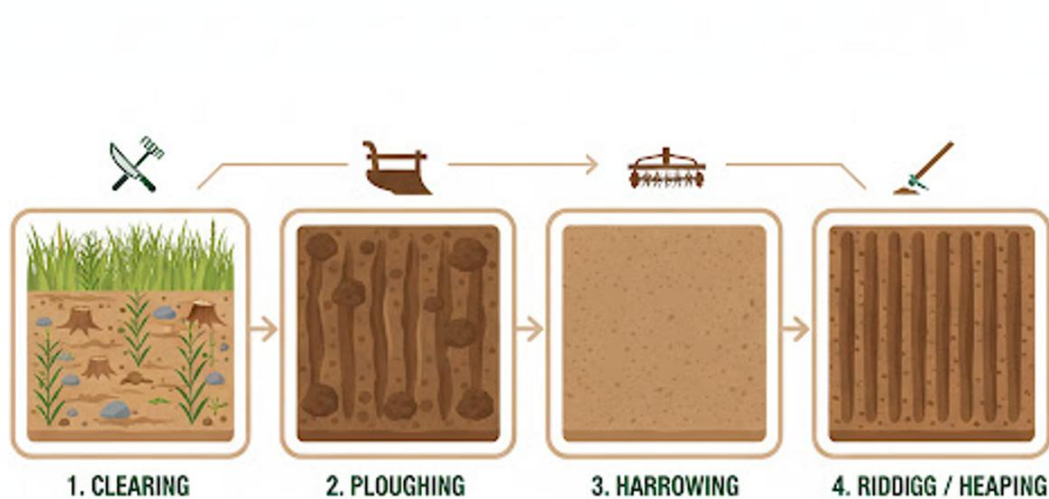
WEEK 4: LAND PREPARATION

🚜 INTRODUCTION TO LAND PREPARATION (TILLAGE)

Land Preparation (also known as **Tillage**) is the series of mechanical operations performed to prepare the soil *before* planting the seeds. It is the "foundation work" for the farm.

You cannot just throw seeds on hard, weedy ground and expect a good crop. You must create a "seedbed" that is soft, weed-free, and has good aeration and drainage.

LAND PREPARATION: A FOUR-STEP PROCESS



OBJECTIVES OF LAND PREPARATION

Why do we do it?

1. **To Destroy Weeds:** This is a primary goal. Tillage kills and buries the existing weeds that would compete with the new crop.
2. **To Loosen the Soil (Improve Tilth):** Hard, compacted soil has no air (poor aeration) and is difficult for roots to penetrate. Tillage "fluffs up" the soil, improving **aeration** and **water infiltration**.

3. **To Create a "Seedbed":** To create a fine, crumbly, smooth surface (a "seedbed") that allows small seeds (like tomato or carrot) to make good contact with the soil and germinate easily.
4. **To Incorporate Organic Matter:** Tillage is how you mix manure, compost, or green manures (cover crops) *into* the soil.
5. **To Create Ridges/Beds:** To shape the land for proper drainage (which is essential in the rainy season).

STAGES OF LAND PREPARATION

These steps must be followed *in order*.

1. CLEARING (LAND CLEARING)

- **Definition:** The *first* operation. This is the removal of the existing vegetation (trees, bushes, grasses, and old crop residues) from the land.
- **Methods:**
 - **Manual:** Using simple hand tools (from 1st Term). (e.g., **Cutlass/Machete** for bushes, **Axe** for trees, **Hoe** for small weeds).
 - *Pros:* Cheap, good for small farms, does not disturb the topsoil.
 - *Cons:* Extremely slow and labor-intensive.
 - **Mechanical:** Using heavy machinery. (e.g., **Bulldozer** (with a rake attachment), **Tractor** with a **Slasher**).
 - *Pros:* Very fast, good for large commercial farms.
 - *Cons:* Very expensive. A bulldozer can *damage* the land by **compacting** the soil and **scrapping away** the fertile topsoil.
 - **Chemical:** Spraying a **non-selective herbicide** (like Glyphosate) to kill all plants before tillage.

2. PLOUGHING (PRIMARY TILLAGE)

- **Definition:** This is the **primary tillage** operation. It is the *first* and *deepest* breaking of the soil.

- **Tool:** A **Plough** (pulled by a tractor or draught animals).
- **Types of Ploughs:**
 - a. **Mouldboard Plough:** This is the classic plough. Its curved blade is designed to **invert the soil** (it *cuts* a slice of soil and *flips it over*). This is excellent for *burying* weeds and manure.
 - b. **Disc Plough:** A heavy-duty plough with large, concave (dished) metal discs. It is better for *hard, dry, or stumpy* soil (common in Nigeria) where a mouldboard plough would break.
- **Result:** Ploughing leaves the soil *very rough*, with large "clods" (lumps).

3. HARROWING (SECONDARY TILLAGE)

- **Definition:** This is the **secondary tillage** operation. It follows ploughing.
- **Purpose:** To **break down the large clods** left by the plough, and to create a *smoother, finer* seedbed.
- **Tool:** A **Harrow** (pulled by a tractor).
- **Types of Harrows:**
 - a. **Disc Harrow:** A set of smaller discs used to cut and break up the clods.
 - b. **Spike-tooth or Spring-tine Harrow:** A "comb-like" tool that is dragged across the soil to smooth and level it, and to pull up any remaining roots.
- **Result:** A fine, crumbly, and level seedbed, ready for planting small seeds (like rice or vegetables).

4. RIDGING (BED FORMATION)

- **Definition:** This is the final shaping of the land *before* planting.
- **Tools:** A **Ridger** (mechanical) or a **Hoe** (manual).
- **Operations:**
 - **Ridges:** Creating a series of raised "peaks" (ridges) and "valleys" (furrows).
 - **Why?** This is the **most important** practice for the rainy season. Crops are planted on the *ridge* (the peak). This provides **excellent drainage**, preventing the roots from sitting in water (waterlogging).
 - **Crops:** Maize, Cassava, Okra.

- **Heaps (Mounds):** Manually creating large, individual mounds of soil.
 - **Why?** This provides a deep, loose, well-drained pile of soil for *large tuber crops*.
 - **Crops: Yams.**
- **Beds:** Creating wide, flat, raised beds (e.g., 1 meter wide).
 - ***Why?** This is the standard for *vegetable nurseries* and for crops that are planted densely (e.g., lettuce, carrots).

EVALUATION

1. What are the 5 main *objectives* (purposes) of land preparation?
2. What is "Primary Tillage"? What is the main tool used for it?
3. What is "Secondary Tillage"? What is the main *purpose* of harrowing?
4. Why is **ridging** so important when farming in the rainy season?
5. What is the difference between a **Mouldboard Plough** and a **Disc Plough**?

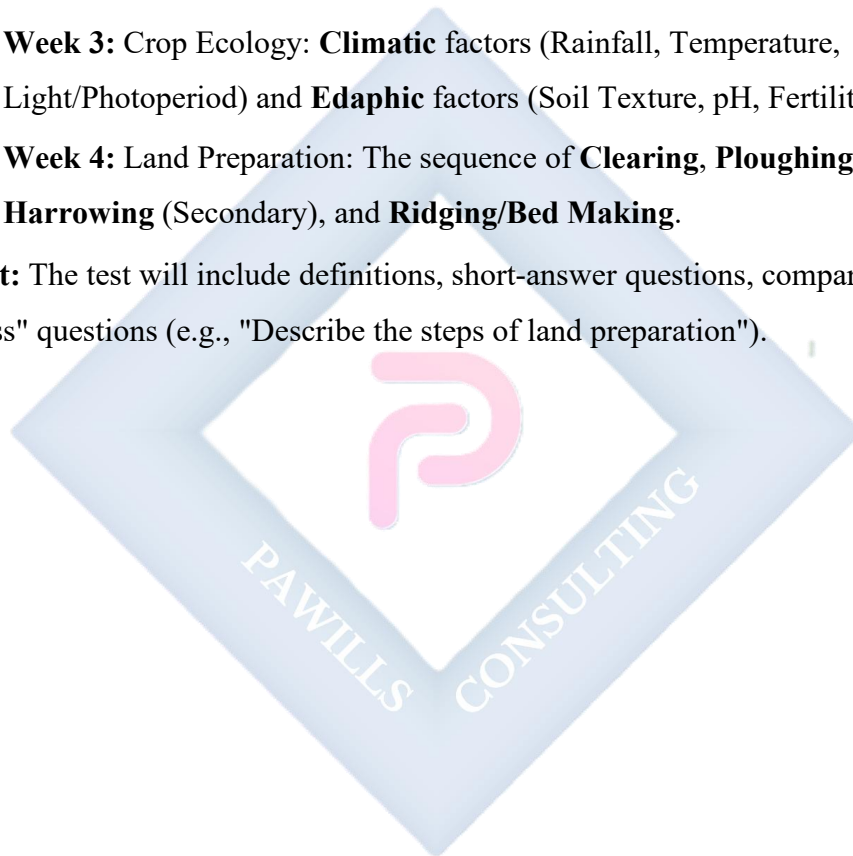
ASSIGNMENT

1. Write a 250-word report explaining the *dangers* of using a bulldozer (a mechanical method) for land clearing. (Hint: topsoil, compaction).
2. Draw simple diagrams of (a) a **Mouldboard Plough**, (b) a **Disc Harrow**, and (c) a **Ridger**.
3. What is the difference between a "Ridge" and a "Heap"? What specific crop is planted on heaps?
4. What is a "seedbed"? Why is a fine seedbed important for small seeds (like carrots)?
5. Define: Tillage, Tilth, Seedbed, Clod, Waterlogging, Compaction.

WEEK 5: MIDTERM EXAMINATION

This week is dedicated to the Midterm Examination, which will cover all topics from Weeks 1–4.

- **Topics:**
 - **Week 1:** Definition and Importance of Crop Production (Food Security, GDP, Jobs).
 - **Week 2:** Classification by Life Cycle: **Annual** (Maize, Rice), **Biennial** (Carrot, Cabbage - and *why* we harvest in Year 1), **Perennial** (Mango, Cocoa).
 - **Week 3:** Crop Ecology: **Climatic** factors (Rainfall, Temperature, Light/Photoperiod) and **Edaphic** factors (Soil Texture, pH, Fertility).
 - **Week 4:** Land Preparation: The sequence of **Clearing**, **Ploughing** (Primary), **Harrowing** (Secondary), and **Ridging/Bed Making**.
- **Format:** The test will include definitions, short-answer questions, comparisons, and "process" questions (e.g., "Describe the steps of land preparation").



WEEK 6: PLANTING OPERATIONS

🌱 INTRODUCTION

Planting is the act of placing the planting material (a seed, seedling, or cutting) into the prepared soil (the seedbed) to germinate and grow.

A successful planting operation depends on three critical factors: **Timing, Spacing, and Depth**. Getting any of these wrong will result in a failed crop.

1. PLANTING TIME (TIMING)

- **Definition:** This is the *most important decision* for a **rain-fed farmer** (a farmer who depends *only* on rainfall).
- **The Rule:** Planting must coincide with the **onset of the rains**.
- **Why?**
 - **If you plant TOO EARLY:** The first rains will come, the seed will germinate, but then a "dry spell" (a 2-week break in the rain) may follow. The young, tender seedlings will *die* from lack of water.
 - **If you plant TOO LATE:** The crop will not have *enough time* to complete its life cycle before the rains stop and the dry season begins. The plant will die before it can produce fruit/seed.
- **Other Factors:**
 - **Market Price:** A smart horticultural farmer with *irrigation* will plant "off-season" (e.g., tomatoes in the dry season) to harvest when the crop is *scarce* and **prices are very high**.
 - **Pest/Disease:** Planting "early" can sometimes avoid the peak of a specific insect pest (like the stem borer in maize).

2. PLANTING DEPTH

- **Definition:** This is the vertical distance *below* the soil surface where the seed is placed.

- **The Rule of Thumb:** A seed should be planted at a depth of **2 to 3 times its own diameter**.
 - *Example:* A small tomato seed (2mm) should be planted 4-6mm deep. A large bean seed (1cm) can be planted 2-3cm deep.
- **Why?**
 - **If you plant TOO SHALLOW:**
 - i. The seed may *dry out* in the sun.
 - ii. It can be *eaten* by birds or ants.
 - iii. It can be *washed away* by heavy rain.
 - **If you plant TOO DEEP:**
 - i. The seed has a *limited* amount of stored energy (in its **cotyledon**).
 - ii. It will use *all* this energy just trying to "push" its way up to the sunlight.
 - iii. It will **run out of energy and die** before it ever reaches the surface. This is a very common cause of "poor germination."

3. PLANTING SPACING (PLANT POPULATION)

- **Definition:** This is the distance *between* one plant and the next (within the row) and the distance *between* the rows.
- **Purpose:** This is a *scientific calculation* to give each plant the *perfect* amount of space to grow without wasting land.
- **The Goal:** To maximize the **plant population** (the number of plants per hectare) to get the *highest yield... without* causing the negative effects of **overcrowding**.
- **Dangers of Overcrowding (Planting TOO CLOSE):**
 - a. **Competition:** The plants will "fight" each other for **light, water, and nutrients**.
 - b. **Weak, Spindly Growth:** Plants will grow tall and "leggy" (spindly) trying to find sunlight. This leads to **lodging** (falling over).
 - c. **Low Yield:** Each individual plant will be weak and produce very small fruit or cobs.

- d. **Disease:** The high humidity and poor airflow in a crowded field are a *perfect* environment for **fungal diseases**.
- **Dangers of Planting TOO FAR:**
 - This is *not* bad for the plant, but it is *bad for the farmer*.
 - It is a **waste of valuable, fertilized land**. The total yield per hectare will be low.
- **Examples of Spacing:**
 - **Maize:** 90cm (between ridges) x 30cm (between plants).
 - **Yams:** 1m x 1m (on heaps).
 - **Cassava:** 1m x 1m.

EVALUATION

1. What are the three critical factors that determine the success of a planting operation?
2. What are the two major risks of planting *too early* in a rain-fed system?
3. What is the "rule of thumb" for planting depth?
4. What will happen to a maize seedling if it is planted 15cm (6 inches) deep?
5. List three negative effects of **overcrowding** (planting too close).

ASSIGNMENT

1. Write a 200-word report explaining *why* planting time is the most important decision for a rain-fed farmer.
2. **Calculation:** A farmer plants maize at a spacing of 100cm x 50cm.
 - (a) How many "stands" (plants) can fit in a plot that is 10m x 10m? (Hint: Convert all units to meters. Area of plot / Area per plant).
3. Define "Plant Population" and explain why it is a balance between "yield per plant" and "yield per hectare."
4. What is "Lodging"?
5. Define: Germination, Cotyledon, Spindly, Rain-fed, Off-season, Etiolation.

WEEK 7: MIDTERM BREAK

This week is for the Midterm Break. There are no formal lesson notes.

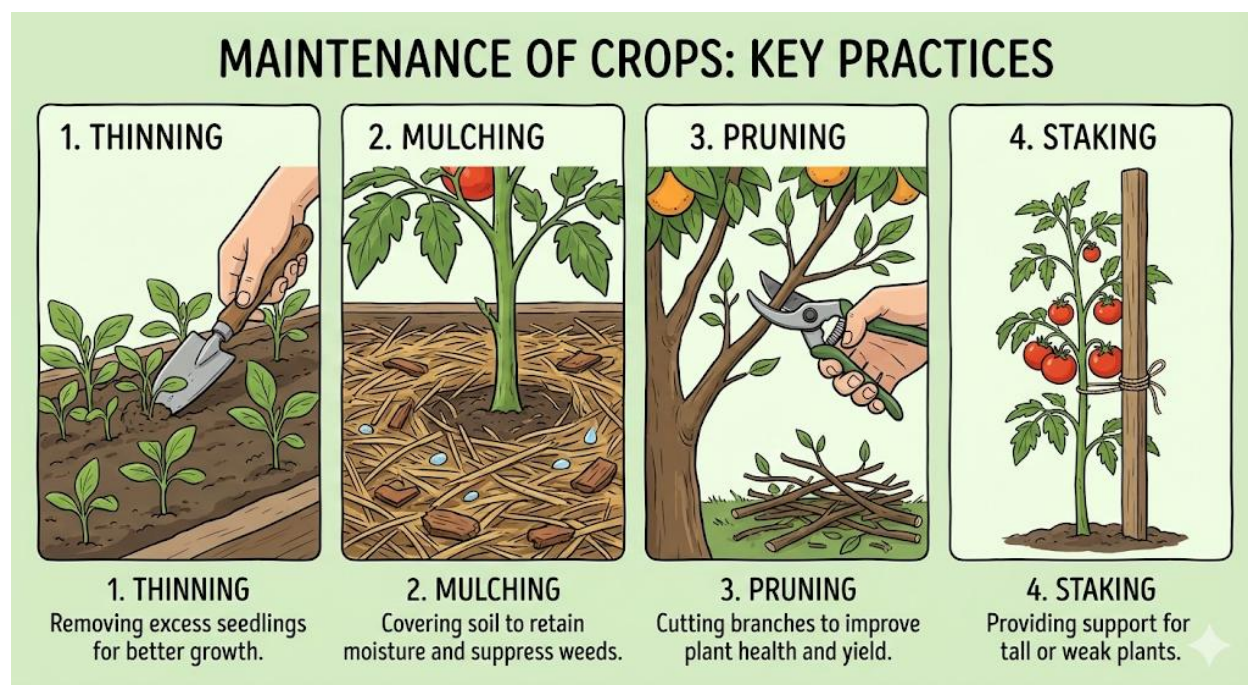


WEEK 8: MAINTENANCE OF CROPS

✿ INTRODUCTION TO POST-PLANTING OPERATIONS

"Maintenance" (also called **Cultural Practices**) refers to all the work done *between* planting and harvesting to care for the growing crops.

Good land preparation and planting can be ruined by poor maintenance. This is the stage where you **protect your investment** by weeding, feeding (Week 1), and supporting the plants.



1. THINNING

- **Definition: Thinning** is the process of *removing* excess or weak seedlings from a stand (a planting hole), to leave only the strongest ones at the correct spacing.
- **Why?**
 - Farmers often plant **2-3 seeds per hole** (e.g., maize, okra, beans). This is an "insurance policy" to make sure at least *one* germinates.
 - If all 3 seeds germinate, they will **overcrowd** each other (as discussed in Week 6), leading to competition for light, water, and nutrients, which results in weak plants and low yield.

- **The Process:**

- **When:** 1-2 weeks after germination, when the seedlings are strong enough to be handled.
- **How:** The farmer inspects each "stand" (hole) and *pulls out* the weaker, smaller seedlings, leaving only the *one* most vigorous and healthy seedling.

2. MULCHING (REVIEW)

- **Definition:** (From Term 1, Week 6). **Mulching** is the practice of *covering* the bare soil surface around the plants with a layer of material.
- **Materials:**
 - **Organic:** Straw, dry grass, wood chips, sawdust.
 - **Synthetic:** Black or silver plastic polythene sheets.
- **The 4 Key Benefits (Review):**
 - a. **Weed Suppression:** Blocks sunlight, so weed seeds cannot germinate.
 - b. **Moisture Conservation:** Acts as a "lid" on the soil, preventing water from evaporating.
 - c. **Temperature Regulation:** Keeps the soil *cooler* in the hot sun.
 - d. **Soil Improvement:** (Organic mulch only). The mulch slowly decomposes, adding *humus* and nutrients to the soil.

3. PRUNING

- **Definition: Pruning** is the *selective* and *purposeful* removal of specific plant parts, such as branches, leaves, flowers, or buds.
- **Purpose:** To *control* the plant's growth and *redirect its energy* to where the farmer wants it (usually to the fruit).
- **Tools:** Secateurs (for small stems), Pruning Saw (for large branches).
- **Examples of Pruning:**
 - **Fruit Trees (Mango, Cocoa):** To create an "**open canopy**" (a V-shape). This allows *air and sunlight* to penetrate the *center* of the tree, which **reduces fungal diseases** and helps all the fruit to ripen.

- **Tomatoes (Suckering):** Indeterminate (vine) tomatoes produce "suckers" (new branches) between the main stem and the leaves. These suckers "steal" energy. The farmer *pinches off* these suckers.
- **Result:** The plant focuses all its energy into producing *fewer* but *much larger* fruits.
- **Cassava:** Removing some stems to encourage tuber growth.

4. STAKING

- **Definition: Staking** is the practice of providing a physical *support* (such as a stake, pole, trellis, or string) for plants that have *weak stems* or a *climbing/vining* growth habit.
- **Why is it Critical?**
 - a. **Disease Prevention (Main Reason):** It keeps the leaves, stems, and fruits *off the wet ground*. This prevents them from *rotting* and stops soil-borne fungal diseases from splashing onto the leaves.
 - b. **Better Sunlight & Airflow:** It exposes all the leaves to the sun and air, which *increases photosynthesis* and *reduces disease*.
 - c. **Better Fruit Quality:** It keeps the fruits (like tomatoes) clean and free from blemishes.
 - d. **Easier Harvest:** It makes the fruits easy to see and pick.
- **CROPS THAT REQUIRE STAKING:**
 - **Yams:** This is *essential* for yams. A staked yam will have a *much higher* yield than an unstaked one.
 - **Climbing Beans**
 - **Indeterminate Tomatoes** (Vine tomatoes)
 - **Cucumber**

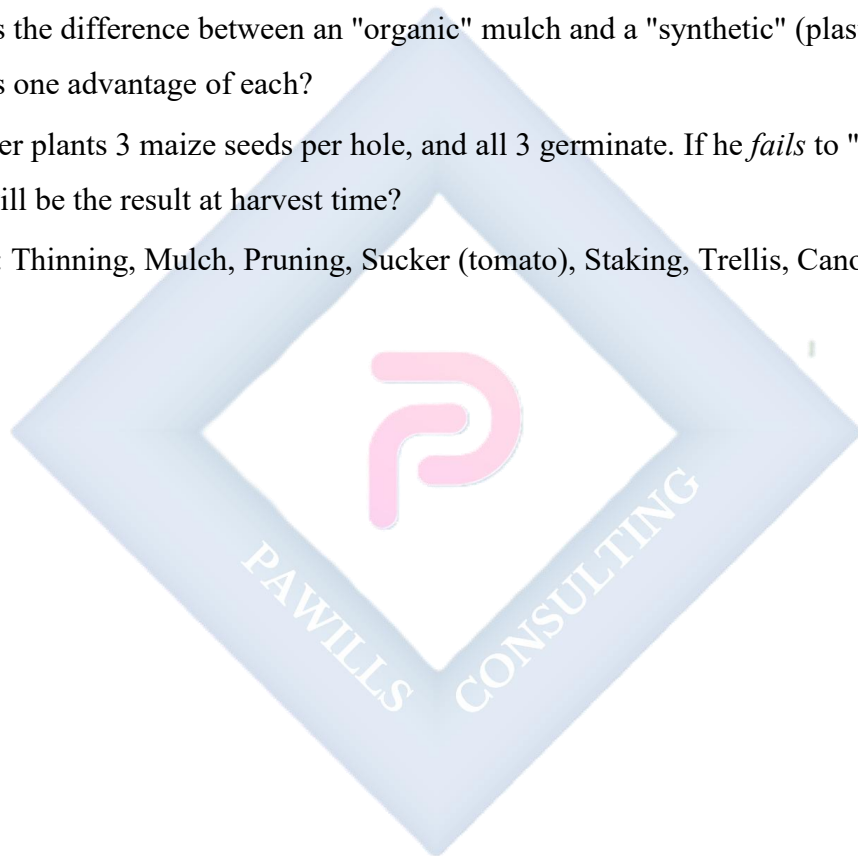
EVALUATION

1. What is "Thinning," and what is the *main purpose* of this operation?
2. List the 4 key benefits of mulching.

3. What is "Pruning"? What is its main goal?
4. Why would a farmer "prune" (sucker) their tomato plants?
5. What is "Staking"? List three reasons why it is a critical practice for yam and tomato.

ASSIGNMENT

1. Write a 200-word report on the **importance of staking for yam production**.
2. Draw a diagram of a **staked tomato plant**. Show the main stake, the plant, and the "ties" holding it.
3. What is the difference between an "organic" mulch and a "synthetic" (plastic) mulch? What is one advantage of each?
4. A farmer plants 3 maize seeds per hole, and all 3 germinate. If he *fails* to "thin" them, what will be the result at harvest time?
5. Define: Thinning, Mulch, Pruning, Sucker (tomato), Staking, Trellis, Canopy, Etiolation.



WEEK 9: CROP PROTECTION

☞ TOPIC: INTEGRATED PEST MANAGEMENT (IPM)



INTRODUCTION

In Term 2, we learned about the "enemies" of the farm: **Weeds, Pests, and Diseases**. We also learned the *individual* control methods (e.g., cultural, chemical, biological).

This week, we learn the *modern strategy* for combining these methods.

The "Old" Way (Over-reliance on Chemicals): The old-fashioned (1960s-1980s) method of crop protection was simple: if you see a bug, spray the *entire* farm with *strong chemicals*

(*pesticides*). This was often done on a "calendar" (e.g., "spray every Monday"), even if there were no pests.

- **The Problems with this:**

- a. **Expensive:** Chemicals cost a lot of money.
- b. **Kills "Good" Insects:** It kills pollinators (like bees) and natural predators (like ladybugs).
- c. **Environmental Pollution:** The poison washes into rivers and soil.
- d. **Pest Resistance:** The few pests that survive (the "strong" ones) breed, and the next generation is *immune* to the spray (pesticide resistance).
- e. **Health Risk:** To the farmer (applicator) and the consumer (residue on food).

INTEGRATED PEST MANAGEMENT (IPM)

DEFINITION: IPM is a *smart, sustainable, and common-sense* strategy for managing pests, weeds, and diseases. It **integrates** multiple control methods (cultural, physical, biological, and chemical) to keep pest populations *below an acceptable level*.

KEY CONCEPTS OF IPM:

- It is *not* about "**Elimination.**" The goal of IPM is *not* to kill *every* single pest. A few pests are not a problem.
- It is about "**Management**" or "**Suppression.**" The goal is to keep the pest population *so low* that it does not cause *economic* damage.
- **Economic Injury Level (EIL):** This is the "tipping point." It is the *lowest pest population* that will cause *economic damage* (i.e., the *cost of the pest damage* becomes *greater* than the *cost of the control method*).
- **Action Threshold:** This is the point *before* the EIL is reached, where the farmer *must* take action to prevent the EIL.

THE IPM PYRAMID (THE STRATEGY)

IPM is a strategy built on a pyramid. A good farmer *always* starts with the foundation (Prevention) and only moves up to the next level if *necessary*. Chemical pesticides are the *small, pointed top*—the **last resort**.

1. FOUNDATION: PREVENTION (CULTURAL CONTROL)

This is the *best, cheapest, and most important* stage. It's about using good farming practices to *prevent* pests and diseases from ever becoming a problem.

- **Crop Rotation:** (The #1 method). Starves soil-borne pests and diseases.
- **Use of Resistant Varieties:** Planting a crop that is *naturally immune* or resistant (e.g., CMD-resistant cassava).
- **Sanitation:** "Roguing" (removing and burning) sick plants. Clearing all old crop debris after harvest to destroy pest eggs.
- **Good Management:** Proper spacing (for airflow), staking (to keep leaves dry), proper irrigation (no waterlogging). A *healthy* plant can fight off pests better.

2. LEVEL 2: PHYSICAL & MECHANICAL CONTROL

If prevention fails and you see pests, the next step is to *physically* remove them.

- **Hand-Picking:** (e.g., picking large grasshoppers or caterpillars off leaves).
- **Traps:** (e.g., **Yellow Sticky Traps** to catch whiteflies, **Pheromone Traps** to lure male insects, Rat traps).
- **Barriers:** (e.g., using **Nets** to keep birds off fruit trees, or a small fence to keep out grasscutters).

3. LEVEL 3: BIOLOGICAL CONTROL

If physical control isn't enough, you use "nature against nature."

- **Importing Predators:** Releasing "good bugs" (like **Ladybugs**) that *eat* the "bad bugs" (like aphids).

- **Biopesticides:** Using natural, non-toxic sprays. (e.g., **Neem Oil** (*Dogonyaro* oil), which suffocates insects and stops them from feeding, or **Bt** (*Bacillus thuringiensis*), a bacteria that kills caterpillars).

4. LEVEL 4: CHEMICAL CONTROL (LAST RESORT)

If (and *only* if) you have monitored the field and the pest population has crossed the **Action Threshold** (and all other methods have failed), you use chemicals.

- **Judicious Use:** You don't "blanket spray." You use the *least toxic* chemical (e.g., a "selective" herbicide, not a non-selective one).
- **Spot Spraying:** You *only* spray the *affected spot* ("hotspot"), not the whole farm. This protects the environment and the "good" insects on the rest of the farm.

EVALUATION

1. What is "IPM" (Integrated Pest Management)?
2. What is the "old" way of pest control, and what are two major problems with it?
3. What is the "Economic Injury Level (EIL)"?
4. What is the "foundation" (the most important level) of the IPM pyramid? Give two examples of practices at this level.
5. What is "Biological Control"? Give one example.

ASSIGNMENT

1. Write a 250-word essay explaining the *strategy* of IPM to a traditional farmer. You must explain *why* it is better, safer, and cheaper than just spraying pesticides.
2. Draw the "IPM Pyramid." Label the four levels (Prevention, Physical, Biological, Chemical).
3. What is a "Resistant Variety"? Why is this the best form of cultural control?
4. Research "**Neem Oil**." How is it made (from the Neem/Dogonyaro tree), and how does it work as a "biopesticide"?
5. Define: IPM, Economic Injury Level, Pesticide Resistance, Biopesticide, Roguing, Pheromone Trap.



WEEK 10: HARVESTING OF FIELD CROPS

INTRODUCTION TO HARVESTING

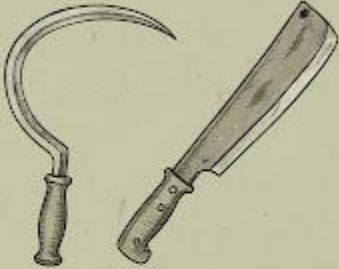
Harvesting is the process of gathering a mature crop from the field. This lesson builds on what we learned in Term 2 (Week 8) about harvesting *horticultural* crops, but now we focus on large-scale **field crops** (agronomic crops).

The key differences are:

- **Horticultural Crops** (Tomatoes, Okra) are often harvested *immature* (Horticultural Maturity) for *freshness, tenderness, and appearance*.
- **Field Crops** (Maize, Rice, Soybeans) are harvested at *full maturity* (Physiological Maturity) for *dry matter, storage, and processing*.

HARVESTING OF FIELD CROPS

1. MANUAL HARVESTING



- **Tools:** Sickle, Cutlass, Hand tools, Human Labor

Pros: Low Initial Cost

- Suitable for Small Farms/Slopes
- Less Crop Damage

Cons:

- High Labor Cost/Time
- Slower Process
- Physically Intensive

2. MECHANICAL HARVESTING



- **Tools:** Combine Harvester, Tractors, Specialized Machinery

Pros: Faster Process

- Covers Large Areas
- Reduced Labor Cost

Cons: High Initial Investment

- High Fuel/Labor Cost
- High Fuel/Maintenance Cost
- Requires Technical Skill

  
Crop Machinery Time/Efficiency

1. TIME OF HARVESTING (MATURITY)

- **Definition:** Field crops must be harvested at **Physiological Maturity**.
- **Physiological Maturity:** This is the stage when the plant has *completed its full life cycle* and the grain or seed is fully developed, has maximum dry weight, and is *starting to dry out*. The plant itself will often be dead or dying (senescence).
- **Harvesting "too early" (High Moisture):**
 - The grain will be soft and have a *high moisture content*.
 - This is the *biggest* risk. This high-moisture grain will **rot and grow mold** (especially **Aflatoxins**) in storage.
- **Harvesting "too late" (Too Dry):**
 - **Shattering:** The seeds (especially in rice and soybeans) become too dry, and the pods "shatter" (burst open), spilling the grain onto the ground before you can harvest it.
 - **Pest Damage:** The crop is exposed in the field for too long, allowing birds, rats, and insects to eat it.
 - **Lodging:** The dry stalks get weak and fall over (lodge).
- **Signs of Physiological Maturity:**
 - **Maize (Corn):** The *leaves and husk* (covering) are brown, dry, and papery. The grain is *hard* (you can't dent it with a fingernail) and has a "black layer" at its base where it attached to the cob.
 - **Rice:** The panicles (seed heads) are heavy, "drooping," and have turned a golden-brown color.
 - **Yams / Cassava:** The leaves turn yellow and begin to fall off (senescence).

2. METHODS OF HARVESTING

This is the main difference between subsistence and commercial farming.

A. MANUAL HARVESTING

- **Definition:** Using manual labor and simple hand tools. This is the most common method in Nigeria.
- **Pros:** Low capital cost (no machines), provides local employment, less damage to crops (if done carefully), can be done on any terrain (e.g., hills).
- **Cons:** Extremely slow, very labor-intensive, high labor cost (if you are hiring).
- **MANUAL TOOLS:**
 - **Cutlass / Machete:** The all-purpose tool. Used for **Yams, Cassava, Sugarcane**, and for cutting **Maize** stalks.
 - **Sickle:** A hand-held tool with a *curved blade*. The *traditional* tool for harvesting grains like **Rice** and **Wheat** (by cutting the stalk).
 - **Hoe / Spade:** Used for digging up **Yams, Cassava**, and **Potatoes**.
 - **Hand Picking:** Used for **Maize** (pulling the cobs), **Cotton** (pulling the bolls), and **Beans** (pulling the dry pods).

B. MECHANICAL HARVESTING

- **Definition:** Using large, engine-powered machinery. This is used in large-scale commercial farming.
- **Pros:** Extremely **fast**. Can harvest many hectares in one day. Low labor cost per hectare.
- **Cons:** **Extremely expensive** (high capital cost of the machine). Requires a *large, flat* field. Can damage the soil (compaction).
- **MECHANICAL TOOLS:**
 - **Combine Harvester:**
 - **Definition:** This is the most advanced harvesting machine. It is a "combine" because it *combines three operations* into one pass over the field.
 - i. **Reaping:** Cuts the standing crop.
 - ii. **Threshing:** Beats the grain out of the seed head/pod.
 - iii. **Winnowing:** Uses a fan to blow the light "chaff" (debris) away, leaving only the clean, heavy grain.

- **Crops: Rice, Maize, Wheat, Soybeans.**
- **Root/Tuber Harvester:** A tractor-pulled implement (like a "digger") that digs under the soil and lifts **Potatoes** or **Cassava** to the surface.
- **Reaper:** A simpler machine that *only* cuts the crop (reaping) and lays it in rows.

EVALUATION

1. What is the main difference between "Horticultural Maturity" and "Physiological Maturity"?
2. What is the main *risk* of harvesting maize when its moisture content is *too high*?
3. What is "Shattering" in rice, and what causes it?
4. What is a **Sickle** used for?
5. What three separate operations are *combined* in a **Combine Harvester**?

ASSIGNMENT

1. Write a 250-word essay comparing **Manual Harvesting** to **Mechanical Harvesting**. Discuss the Pros and Cons of each (cost, speed, labor).
2. Draw a simple diagram of a **Sickle** and a **Cutlass**. Label them and state their primary use in harvesting.
3. What is "Threshing"? What is "Winnowing"? (These are two of the functions of a combine harvester).
4. Research the harvesting of **Cassava**. Describe the manual process step-by-step.
5. Define: Physiological Maturity, Senescence, Shattering, Lodging, Threshing, Winnowing, Chaff.

WEEK 11: REVISION

GENERAL REVIEW OF ALL THIRD TERM TOPICS

This week is for a comprehensive review of all topics covered this term. We have covered the entire life cycle of a crop, from land preparation to harvesting, and the management practices needed to ensure success.

- **WEEK 1: CROP PRODUCTION**

- **Definition:** The cultivation of plants (Horticulture + **Agronomy**).
- **Importance: Food Security** (staples: yam, cassava, rice, maize), **Economy** (Jobs, GDP, Foreign Exchange), **Raw Materials** (textiles, food processing).

- **WEEK 2: LIFE CYCLES**

- **Annuals:** One season, seed-to-seed, die. (e.g., Maize, Rice, Tomato).
- **Biennials:** Two seasons. Year 1 (vegetative/storage), Year 2 (flower/seed). (e.g., Carrot, Cabbage).
- **Perennials:** Live for many years. (e.g., Mango, Cocoa, Oil Palm, Cassava, Plantain).

- **WEEK 3: CROP ECOLOGY**

- The study of a crop's environmental needs.
- **Climatic: Rainfall** (amount/distribution), **Temperature** (Tropical vs. Temperate - Jos Plateau), **Light (Photosynthesis, Photoperiodism)**.
- **Edaphic (Soil): Texture** (Sandy-loam for yams, Clay for rice), **pH** (acidity - needs **Liming** in the South), **Fertility (Legumes fix Nitrogen)**.

- **WEEK 4: LAND PREPARATION**

- The *order* of operations:
 - a. **Clearing:** (Manual or Mechanical).
 - b. **Ploughing:** (Primary Tillage). Inverts soil, buries weeds (Mouldboard vs. Disc Plough).
 - c. **Harrowing:** (Secondary Tillage). Breaks clods, creates fine **seedbed**.

- d. **Ridging/Heaping:** (Bed formation). For **drainage** (ridges) and tuber expansion (heaps for yams).
- **WEEK 6: PLANTING OPERATIONS**
 - **Timing:** Critical for rain-fed. Must match the *onset of rains* (not too early, not too late).
 - **Depth:** Rule of thumb = 2-3x seed diameter. (Too deep = seedling runs out of energy/cotyledon).
 - **Spacing:** A balance to avoid **overcrowding** (which causes competition, lodging, and disease).
- **WEEK 8: CROP MAINTENANCE**
 - **Thinning:** Removing weak seedlings to reduce competition (when 2-3 seeds are planted per hole).
 - **Mulching:** Covering soil (conserves water, stops weeds, regulates temp).
 - **Pruning:** Removing parts to redirect energy (e.g., tomato suckers).
 - **Staking:** Supporting weak-stemmed plants (Yams, Tomatoes) to prevent rot and disease.
- **WEEK 9: CROP PROTECTION (IPM)**
 - **IPM (Integrated Pest Management):** A *smart strategy* (not just chemicals) to keep pests below the **Economic Injury Level (EIL)**.
 - **Pyramid:** (1) **Prevention/Cultural** (Rotation, Resistant Varieties, Sanitation), (2) **Physical** (Traps, Hand-picking), (3) **Biological** (Ladybugs, Neem Oil), (4) **Chemical** (Pesticides, as a *last resort*).
- **WEEK 10: HARVESTING FIELD CROPS**
 - **Maturity:** Must be **Physiological Maturity** (fully developed and *dry*).
 - **Risks:** Too early (rot in storage), Too late (**Shattering**, pest damage, lodging).
 - **Tools (Manual):** Cutlass (Yam), **Sickle** (Rice), Hoe (Cassava).
 - **Tools (Mechanical):** **Combine Harvester** (Reaps, Threshes, and Winnows grains).

EVALUATION

1. What is the main difference between Pomology and Olericulture?
2. What is the single most important soil management practice, and how does it help *both* sandy and clay soils?
3. What is "Hardening Off," and why is it essential?
4. What are the two main *advantages* of Asexual (Vegetative) propagation over Sexual (Seed) propagation?
5. What is "Fertigation," and which irrigation method makes it possible?

ASSIGNMENT

1. **Project:** You are a small farmer in Nigeria. You want to grow one hectare of high-quality **tomatoes** for sale in the *dry season*. Write a one-page "Production Plan" that addresses:
 - (a) How you will propagate the plants (Week 9).
 - (b) How you will raise the young plants (Week 8).
 - (c) How you will manage the soil's pH and fertility (Week 6).
 - (d) How you will water your crop (Week 10).
 - (e) The tools you will need (Week 4).
2. Draw a simple diagram of "Grafting," showing and labeling the **Scion** and the **Rootstock**.
3. Create a table comparing the 3 irrigation methods (Surface, Sprinkler, Drip) based on **Cost, Water Efficiency, and Disease Risk**.
4. Define the 10 most important terms you learned this term.
5. What is the difference between a "Cash Crop" and a "Food Crop"? Give an example of a crop that is both.